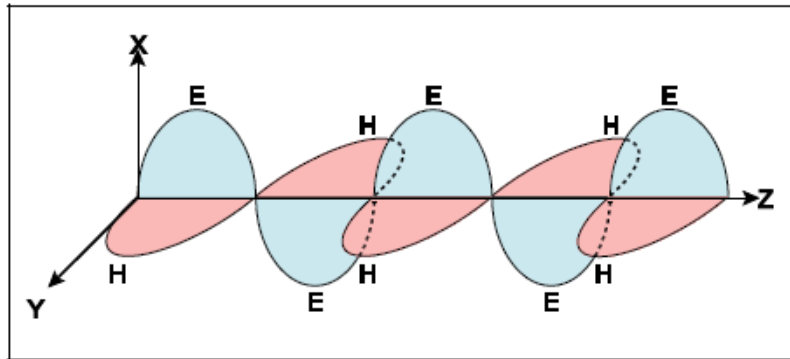


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Radio Wave

If an electric current is passed through a wire an oscillating current will setup in the wire. The resulting wave is made up of two components, an electrical (E) field and a magnetic (H) field. Together they will be called Electromagnetic wave or Radio Wave. Speed of radio wave is same as speed of light which is 3×10^8 m/s or 162 000 nautical miles per second.



Components of Radio Wave -

- Frequency is the number of cycles per second. It is denoted in Hertz.
 Kilo-Hertz (kHz) = 10^3 Hz = 1 000 Hz
 Mega-Hertz (MHz) = 10^6 Hz = 1 000 000 Hz
 Giga-Hertz (GHz) = 10^9 Hz = 1 000 000 000 Hz
- Wavelength is the physical distance travelled by the radio wave or distance between successive crest and successive trough.

Relationship between frequency and wavelength -

$$f = \frac{c}{\lambda} \text{ where } c = 3 \times 10^8 \text{ m/s.}$$

- Amplitude – It is the maximum displacement attained by the wave from its mean value.

Polarisation –

- The polarisation of radio waves is defined as the plane of the electric field and is dependent on the plane of the aerial.
- If a wave is vertically polarised then electric component travels in the vertical plane and the associated magnetic component in the horizontal plane
- If a wave is horizontally polarised then electric component travels in the horizontal plane and the associated magnetic component in the vertical plane.

Frequency Bands –

Frequency Band	Frequencies	Wavelengths	Civil Aeronautical Usage
VLF	3 – 30 kHz	100 – 10 km	Nil
LF	30 – 300 kHz	10 – 1 km	NDB/ADF, LORAN C
MF	300 – 3000 kHz	1000 – 100 m	NDB/ADF, long range communications
HF	3 – 30 MHz	100 – 10 m	Long range communications
VHF	30 – 300 MHz	10 – 1 m	Short range communication, VDF, VOR, ILS localiser, marker beacons
UHF	300 – 3000 MHz	100 – 10 cm	ILS glidepath, DME, SSR, Satellite communications, GNSS, long range radar
SHF	3 – 30 GHz	10 – 1 cm	RADALT, AWR, MLS, short range radars
EHF	30 – 300 GHz	10 – 1 mm	Nil

Other Bands –

L Band	1 – 2 GHz
S Band	2 – 4 GHz
C Band	4 - 8 GHz
X Band	8 – 12.5 GHz

Radio Propagation Theory

Propagation is the movement of the radio waves through the atmosphere.

Propagation Paths –

Surface wave Propagation (Range) :-

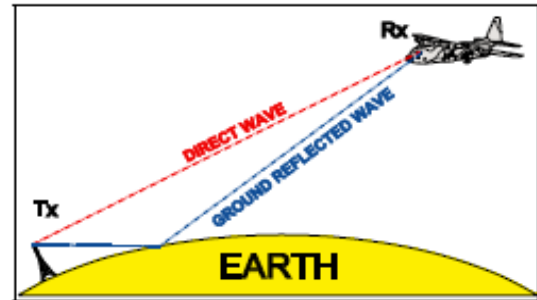
$$\text{Sea} = 3 \times \sqrt{(\text{Power of Tx})}$$

$$\text{Land} = 2 \times \sqrt{(\text{Power of Tx})}$$

a. **Surface Waves** – They travel along the surface of the earth and bend around the surface of the earth.

b. **Space Wave** – It is made up of two paths, a direct wave and a reflected wave.

- i. **Direct Wave** – Waves reaching receiver in straight line (line of sight) are called direct waves.
- ii. **Reflected Waves** – Waves reflected by an object.



c. **Sky Wave** – Waves refracted by the ionosphere are called Sky waves and are used in Long Range Navigation.

Note – Space Wave and Surface Wave combine to form **Ground Waves**.

Propagation of Surface Waves – Radio waves do not follow optical path, they curve slightly which increases their range. This is due to –

1. **Diffraction** - The process in which waves bend around small obstacles or spread out after passing through small openings. It depends on frequency, lower the frequency higher the diffraction.
2. **Attenuation** - Attenuation is the term given to the loss of signal strength in a radio wave as it travels outward from the transmitter. Attenuation depends on –
 - Type of surface – More range over sea than land due to less attenuation over sea.
 - Frequency in use - Higher the frequency, greater the attenuation.
 - Polarisation – Vertically polarized waves have less attenuation and more range.

$$\text{Freq} \propto \text{Attenuation}$$

Propagation of Sky Waves –

$$\text{Freq} \propto \frac{1}{\text{Range}}$$

Ionosphere - The layer extending from about 60 km till atmosphere is ionosphere. It consists of electrons emitted from UV rays which come from sun. Because absorption of solar radiation is uneven, the ionosphere exists in 3 layers –

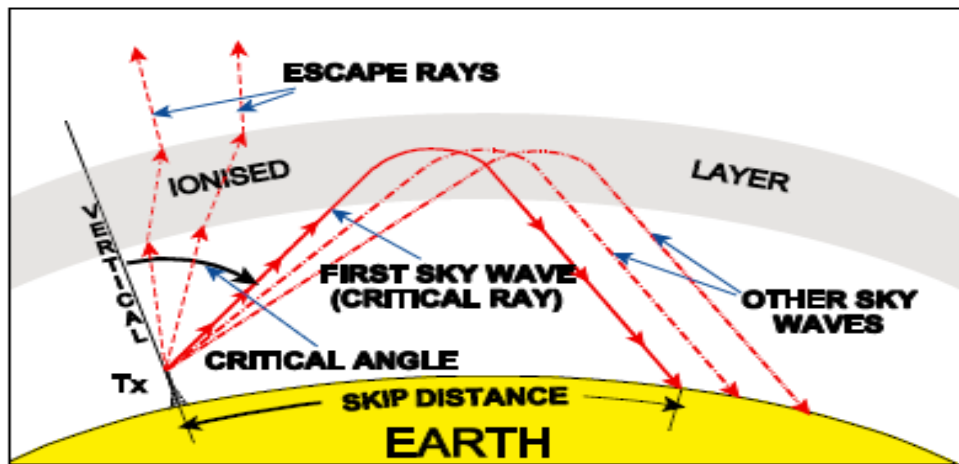
D Layer - 50 – 100km (Avg 75km)

E Layer – 100-150km (Avg 125km). It is also called Kennelly Heaviside layer.

F Layer – 150 -350km (avg 225km). It is also called Appleton layer. This often exists as F1 and F2 layer. Has maximum Variations.

Diurnal Activity – Level of ionization will be more during daytime. At night, D layer vanishes. E layer reduces and it rises. F1 & F2 combine to form one layer and height increases.

$$\text{Space wave Propagation(Range)} = 1.25 \times (\sqrt{\text{ht. of Tx}} + \sqrt{\text{ht. of Rx}})$$

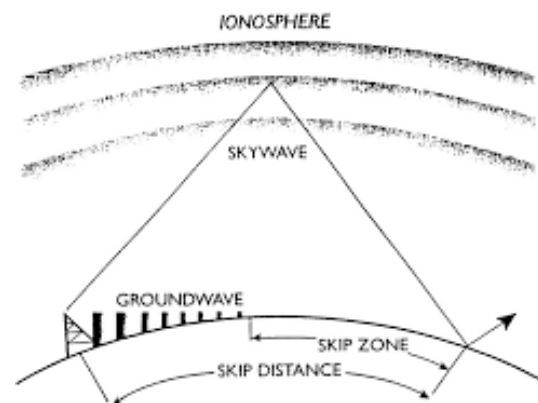


Critical Ray - A signal entering ionosphere will undergo refraction. An angle is reached where total internal refraction occurs and the wave returns to the surface is known as the first returning skywave or Critical Ray.

Critical Angle - The angle measured from the transmitter and the first sky wave is known as the critical angle.

Skip Distance - The distance from the transmitter to the point where the first returning skywave appears at the surface is known as the skip distance.

Dead Space – Distance from the point where the surface wave is totally attenuated to the point where the first returning skywave appears. There will be no detectable signal in this region.



Frequency ↑ : Dead zone ↑ : Skip Dist. ↑

Power ↑ : Dead zone ↓ : Skip Dist. ↑

Multi Hop Refraction – If returning signal is strong, it will be reflected back to ionosphere again. This may happen several time.

Fading - Two signal from same transmitter arriving at same receiver following different path may be out of phase causing fading. It is generally caused during night.

Note - Sky wave is present in VLF, LF, MF and HF but used only for HF Communication not navigation.

VHF Range Formula – $R = 1.25\sqrt{H_{TX}} + 1.25\sqrt{H_{RX}}$

The average maximum range for refraction from E Layer at 125km is 1350nm and the average maximum range from the F Layer at 225 km is 2200nm.

Factors affecting VHF Range –

- i. Height of Receiver
- ii. Height of Transmitter
- iii. Transmission Power
- iv. Obstacle near the transmission site.
- v. Fading caused as a/c may receive direct and ground reflected waves.

Super-Refraction (Duct Propagation) -

It occurs at frequency range of VHF and above. Radio waves experience greater refraction, and they bent downwards towards the earth's surface more than in normal conditions, giving more range. The conditions which give rise to super-refraction are:

- Temperature falling more slowly with height than standard (or Inversion).
- Fine weather and high pressure systems. (Stable Conditions)
- Warm air flowing over a cooler surfaces.

Frequency Band	Propagation Path
VLF	Ducting
LF	Surface Wave (Skywave)
MF	Surface Wave (Skywave)
HF	Skywave (Surface Wave)
VHF	Space Wave
UHF	Space Wave
SHF	Space Wave
EHF	Space Wave

SELCAL (Selective Calling System) – The system relieves pilot of the tiresome task of maintaining a continuous listening watch. When the ground station intends to contact an aircraft maintaining SELCAL, it transmits two coded tone pulses. This activates a cockpit call system by flashing a lamp or ringing a bell.

ACARS (Aircraft Communication Addressing and Reporting System) – It is a common data link between the airline ground operations base and the aircraft.

SATCOM – Satellite Communication through INMARSAT. These are a group of geostationary satellites provided by international maritime satellite organization. It transmits in L Band and provides communication between 80°N and 80°S.

CPDLC - Controller- Pilot Data Link Communication. It is a method by which air traffic controllers can communicate with the pilot over a datalink system.

Modulation

Modulation is the process of superimposing intelligence to a radio wave. In this process radio wave acts as a carrier wave and the intelligence act as modulating wave.

Types of Modulation are:-

- a. **Keying** – Information is added by interrupting the wave by giving short and long bursts of energy. Eg- Telegraphy.
- b. **Amplitude Modulation (AM)** - In AM the amplitude of the audio frequency (AF) modifies the amplitude of the radio frequency (RF). (Frequency Constant)
- c. **Frequency Modulation (FM)** - In FM, the amplitude of the audio frequency modifies the frequency of the carrier wave. (Amplitude Constant)
- d. **Pulse Modulation** – The Modulating pulse amplitude modulates the carrier, giving it shape of a pulse. Used in Radar Systems.
- e. **Phase Modulation** - The phase of the carrier wave is modified by the input signal. It is used in VOR.

SINGLE SIDEBAND (SSB)

It is a refinement of AM, it uses transmitter power and bandwidth more efficiently.

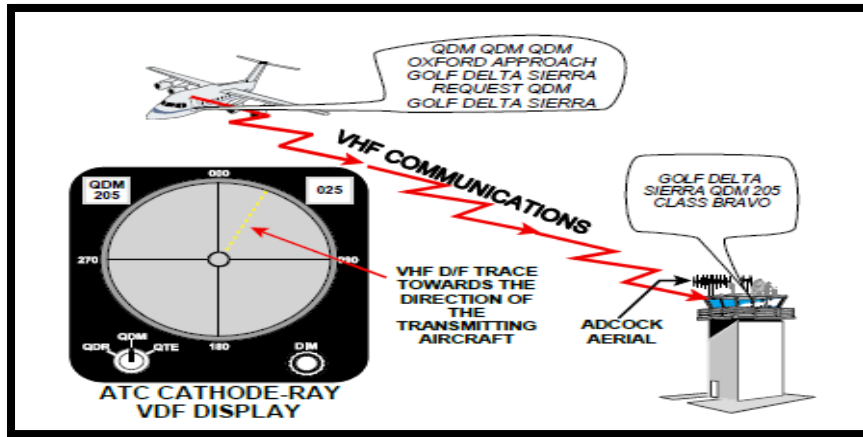
The main advantages of SSB are:

- Double the number of channels available.
- Better signal/noise ratio (less interference).
- Less power required hence lighter equipment.

Emission Designators – They comprise 3 alphanumerics, where the first letter defines the nature of the modulation, the second digit the nature of the signal used for the modulation and the third letter the type of information carried. Some examples are -

ADF	N0NA1A or N0NA2A
VHF	A3E
HF	J3E
VOR	A9W
ILS	A8W
Marker Beacons	A2A
DME	P0N

VHF Direction Finder (VDF)



It is a ground station which measures bearing of an aircraft's VHF transmissions (118 - 137 MHz, emission code A3E). The ground installation comprises an adcock aerial and a CRT display.

Q Codes -

QDM - Aircraft's Magnetic Heading To the station in nil winds.

QDR - Aircraft's Magnetic Bearing From the station.

QTE - Aircraft's True Bearing From the station.

QUJ - Aircraft's True Track To the station.

VDF SUMMARY -

Bearings	QDM - Mag TO station QDR - Mag FROM station QTE - True FROM station
Uses	Check track, Position line, Homing, Let-downs
Class	A = ± 2 B = ± 5 C = ± 10 D = >10
Principle	Ground equipment - Adcock aerial, CRT
Range	Line of sight, Power of transmitters, Intervening high ground, Atmospheric conditions (ducting)
Accuracy	Propagation error, site error, Aircraft attitude, Overhead station, Fading due to multi-path signals
Emergency service	Position fixing by auto-triangulation
Let-down service	QDM procedure - pilot interpreted QGH procedure - controller interpreted

NDB/ADF

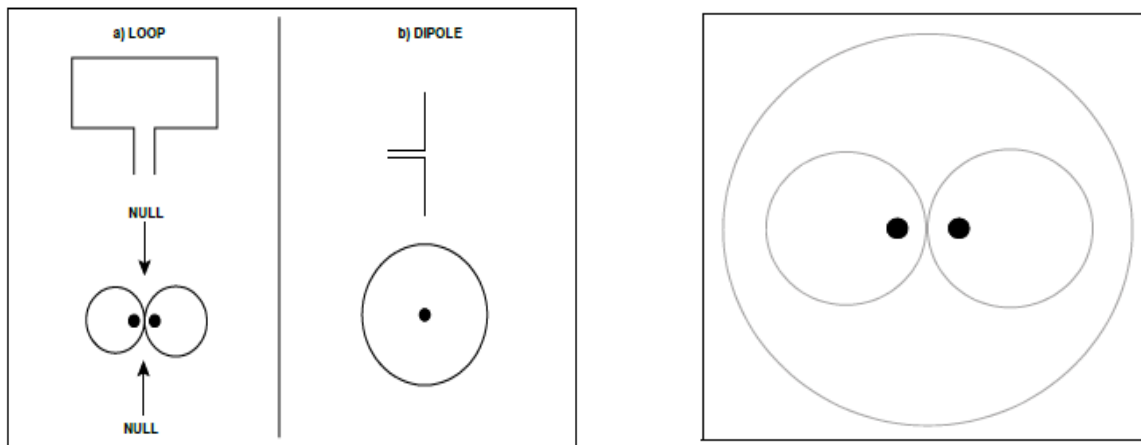


- The Non Directional Beacon (NDB) is a ground based transmitter which transmits vertically polarised radio signals, in all directions in the LF and MF bands.
- Automatic Direction Finder (ADF) is an airborne equipment used in conjunction with a (NDB) on the ground to measure the direction of incoming signal.
- ADF measures direction in relation to the nose of the aircraft. Hence, they are Relative Bearing.
- Range is 200 nm by day and 70 nm by night.
- Accuracy is $\pm 5^\circ$ within beacon protected range.
- Frequency used is 190 – 1750 kHz (Upper LF & Lower MF).

Principle of Operation –

- NDB/ADF uses vertically polarized signals.
- The ADF measures the bearing of a NDB relative to the fore/aft axis of the aircraft.
- If a loop aerial is placed in the plane of the transmitted radio frequency a voltage will be generated in the vertical elements of the loop because of the phase difference of the wave in each of the vertical elements.
- When the loop is parallel to direction of incoming signals maximum phase difference and maximum current is induced.
- When both members are perpendicular to each other, both will receive incoming signal together at same phase and current flow will be nil.
- If the loop is further rotated, current flows in reverse direction.
- The polar diagram formed is a figure of eight. There are two null positions.
- Two minimas and two maximas cause 180° ambiguity.
- To resolve this ambiguity a simple dipole aerial, called a sense aerial, is added.

- The polar diagram of the sense aerial is circular.
- When combined the resultant polar diagram is known as a CARDIOID. The cardioid has a single null.



Types of NDB –

- Locator NDB** - These are low powered NDBs used for airfield or runway approach procedures or are co-located with, the outer and middle markers of an ILS system. They normally have ranges of 10 to 25nm.
- Homing and Holding NDB** – Used as approach aid and holding aid in the vicinity of an airfield. Rated coverage of 50nm.
- En route NDBs** - These have a range of more than 50nm which goes up to 200nm in oceanic areas. They are used for en route and airways navigation.

Coverage of NDB Transmissions – ICAO Recommends 3:1 protection ratio (wanted signal: unwanted signal). The rated coverage of NDB depends upon:-

- Frequency
- Transmission Power
- Conductivity of path to the coverage boundary.

Emission Characteristics - The NDBs have a 2 or 3 letter identification and there are two types of emission: **N0NA1A** and **N0NA2A**

BEAT FREQUENCY OSCILLATOR (BFO)-

- The N0N part of the emission is the transmission of an unmodulated carrier wave, hence, no intelligence.
- The frequency used by ADF is so high that it is not audible.
- A BFO is provided on ADF equipment to make the signal audible.
- When selected, the BFO produces an offset frequency within the receiver which when combined with the received frequency produces a tone of audible signals say 400 or 1,020 Hz.
- BFO is used as follows –

	Tuning	Ident	Monitoring
NONA1A	ON	ON	ON
NONA2A	ON	OFF	OFF

Uses of the NDB -

- En-route navigational bearings.
- Homing to or flying from the NDB when maintaining airway centre-lines.
- Holding overhead at an assigned level in a race-track pattern.
- Runway instrument approach procedures.

Homing on to a Station – (Thumb rule)

Winds from starboard – $ADF < 360^\circ$

Winds from port – $ADF > 360^\circ$

Tracking Out – In Nil winds ADF is 180° . In winds-

Starboard Drift – $ADF = 180 + \text{Starboard Drift}$

Port Drift – $ADF = 180 - \text{Port Drift}$

FACTORS AFFECTING ADF ACCURACY

- Designated Operational Coverage (DOC)** - The DOC of NDBs is based upon a daytime protection ratio (signal/noise ratio of 3:1) between wanted and unwanted signals that permits the required level of bearing accuracy.
- Precipitation static** - It causes a reduction in the signal:noise ratio which affects the accuracy of the bearings and can, in extreme circumstances completely mask the incoming signal.
- Thunderstorms** - Thunderstorms discharges cause bearing errors in the ADF. The needle will move rapidly to point to the Cb. When there are several active cells close together, it is possible for the needle to point to them for prolonged periods. Due to this reason, Thunderstorm error is the biggest error in NDB.
- Night Effect** - At night the D-region disappears allowing skywave contamination of the surface wave being used. This arises for two reasons: phase interference of the skywave with the surface wave because of the different paths and the induction of currents in the horizontal elements of the loop aerial. The effect becomes significant at 70 – 100 nm from the NDB. The error is worst around dawn and dusk, when the ionosphere is in transition. It can be reduced by: -
 - ✓ Use beacon with greater power output.
 - ✓ Use beacon with lower frequency.
 - ✓ Use beacon closer to aircraft.
 - ✓ Avoid use of the equipment within 1 hour of sunrise or sunset.
- Station Interference** – If 2 stations are on same frequency, ADF needle will lock in between as a resultant.

- vi. **Mountain Effect** - Mountainous areas can cause reflections and diffraction of the transmitted radio waves to produce errors in ADF systems.
- vii. **Coastal Refraction** - Radio waves speed up over water due to the reduced absorption of energy (attenuation) compared to that which occurs over land. This speeding up causes the wave front to bend (Refract) away from its normal path and pull it towards the coast. Refraction is negligible at 90° to the coast but increases as the angle of incidence increases. The effect can be minimised by:
 - ✓ Using NDBs on or near to the coast.
 - ✓ Flying higher.
 - ✓ Using signals that cross the coast at or near to 90°.
- viii. **Quadrantal Error** - Incoming NDB signals are refracted towards the fore and aft airframe axis. The maximum refraction occurs in the quadrants (i.e on relative bearings of 045°, 135°, 225° & 315°). Older ADF systems are regularly 'swung' to assess the value of quadrantal error. In modern aircraft the error is determined by the manufacturer and corrections are put into the equipment to reduce the effect to a minimum.

Factors Affecting ADF Range –

1. **NDB transmission power** - To double the NDB range, quadruple the power output of the transmitter.
2. **Type of surface** – More range over water.
 $3 \times \sqrt{P}$ (W) over water
 $2 \times \sqrt{P}$ (W) over land
3. **Frequency** - The lower the frequency, the greater the surface wave (greater diffraction, lower attenuation).
4. **Type of Emission** - Non A1A NDBs have greater ranges than non A2A.

ACCURACY

The accuracy of ADF is +/-5° within the designated operational coverage, by day only.

ADF SUMMARY –

NDB	Ground transmitter in LF or MF band (190 - 1750 kHz)
Types of NDB	Locator (L) - airfield let-down (10 - 25 nm) En Route - Nav-aid (50 nm or more)
Range (nm)	$3 \times \sqrt{P}$ (W) over water $2 \times \sqrt{P}$ (W) over land
ADF	Airborne equipment - aerals, receiver, control unit, indicator (RBI / RMI)
Principle of operation	(Relative) Bearing by switched cardioids
Frequencies	190 - 1750 kHz (LF & MF)
Emission characteristics	N0NA1A - BFO ON for tuning, identification and monitoring N0NA2A - BFO ON for tuning, OFF otherwise
Presentation	RBI or RMI
Uses of NDB	Homing, Holding, Approach, En route nav-aid
Errors	Static interference (precipitation and thunderstorms) Station interference Night effect Mountain effect Coastal refraction Quadrantal error Bank angle (dip) Lack of failure warning
Accuracy	(Day Only) +/- 5° within the DOC

VHF Omni-Directional Range (VOR)

Introduction –

- The VHF Omni-directional Range (VOR) is the standard short range navigation aid.
- It produces 360 radials/tracks at 1° spacing which are aligned in relation to magnetic north at the VOR location.
- It is practically free from static interference and is not affected by sky-waves, which enables it to be used day and night.
- When the VOR frequency is paired with a co-located Distance Measuring Equipment (DME) an instantaneous range and bearing fix is obtained.
- The equipment operates within the frequency range of **108 - 117.95MHz**.
- Band 108-112 MHz shared with ILS so uses only even first decimals.
- In band 112-117.95MHz all frequencies are assigned for VOR.
- The emission characteristics are **A9W**.
- VOR has the following uses:
 - ✓ Marking the beginning, the end and centre-line of airways or sections of airways.
 - ✓ As a let-down aid at airfields using published procedures.
 - ✓ As a holding point for aircraft.
 - ✓ As a source of en-route navigational position lines.

Advantages over ADF –

- VOR is free from static and does not suffer night effect
- Cockpit indications are easier to see and interpret.
- Coverage ranges are more accurate.
- Warning device in VOR to tell if instrument is reliable or not.
- VOR can be used for RNAV and can be coupled with autopilot.

Radial – It is the magnetic bearing of the aircraft from the stations. Tail of VOR needle always indicate radial. Radial does not change with heading of the aircraft.

Principle of Operation –

- VOR bearing is obtained by phase comparison.
- The ground beacon has two aerials – One fixed and one rotating.
- The fixed aerial produces a 30Hz frequency modulated omni-directional, reference signal which produces constant phase regardless of a receiver's bearing from the VOR.
- The polar diagram of reference signal is a circle.
- The rotating aerial produces a 30Hz amplitude modulated variable phase (directional) signal created by the rotating transmission pattern.
- The polar diagram of variable signal is rotating figure of 8.

- Phase of received signal will vary with the position of the aircraft (receiver).
- Both signals are in same phase when receiver is in magnetic north from transmitter.
- Phase difference between the two gives magnetic bearing of aircraft from the station (Radial).
- The net effect of these two polar diagrams (circle & rotating figure of 8) is almost like a cardioid but without a null. It is called Limacon.
- When the variable signal is phase compared with the reference signal, there would be a phase difference. This is arranged in a manner that phase difference is 0° at magnetic north, 90° at 090 and so on.
- The phase difference measured at any point will equate to the aircrafts magnetic bearing from the VOR.
- Eg- A phase diff. of 227° measured at the aircraft = 227° Radial.

Identification –

- VORs use 3 letter aural morse sent at approximately every 10 seconds.
- A continuous tone or a series of dots identifies a TEST VOR (VOT)

Difference between Cardioid and Limacon –

1. Limacon forms on ground while cardioid is on aircraft.
2. At null position, cardioid has zero reception and limacon has same reception.
3. Cardioid is fixed while limacon is rotating.

Types of VOR –

- i. **BVOR** - A broadcast VOR which gives weather and airfield information between beacon identification.
- ii. **DVOR** - A Doppler VOR - this overcomes siting errors.
- iii. **TVOR** - Terminal VOR which has only low power; and is used at major airfields.
- iv. **VOT** - This is found at certain airfields and broadcasts a fixed omni-directional signal for a 360° test radial. This is not for navigation use but is used to test an aircraft's equipment accuracy before IFR flight. More than $\pm 4^\circ$ indicates that equipment needs servicing.
- v. **VORTAC** - Co-located VOR and TACAN (DME) beacons.

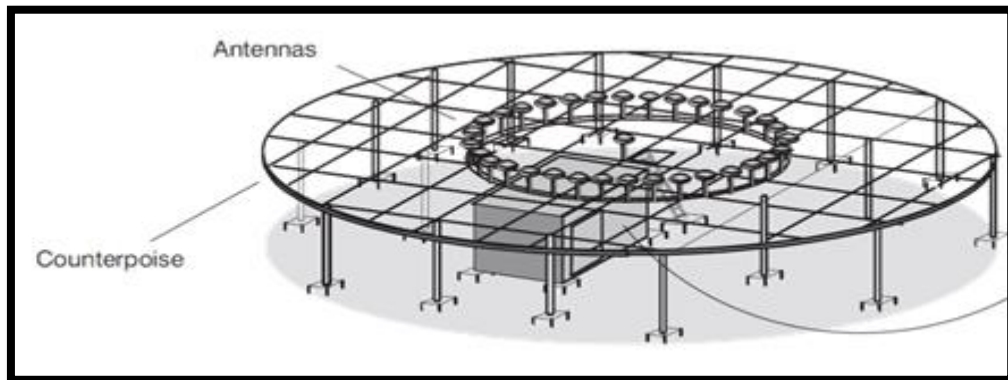
Doppler VOR (DVOR) -

Doppler VORs are second generation VORs. Although their transmission frequencies are the same, the transmitted bearing accuracy is improved as the transmissions are less sensitive to site error.

The transmission differences are:

- Modulation is reversed (RAM – Reference AM).
- The reference signal is AM.
- The variable phase directional signal is FM.

- There is no rotating aerial.
- It is a set of 50 aerials encircling a central aerial at a diameter of 44ft.
- The central aerial transmits the reference signal which is 30Hz AM.
- The 50 fixed aerials in combination transmit variable signal which is 30 Hz FM.
- The combined transmission of all 51 aerials produces a rotating anti-clockwise limaçon.
- Frequency is same and the airborne equipment is same, there is no difference to the pilot.



Factors affecting VOR Range –

- a. **Transmitter Power** - The higher the transmitter power, the greater the range.
- b. **The transmitter and receiver height** - As VOR transmissions give line of sight ranges.

$$\text{Maximum theoretical reception range (nm)} = 1.25 \times (\sqrt{H1} + \sqrt{H2})$$

Where: H1 = Receiver height in feet amsl.

H2 = Transmitter height in feet amsl.

- c. **Nature of terrain** - Uneven terrain, intervening high ground, mountains, man-made structures etc., cause VOR bearings to be stopped (screened), reflected, or bent (scalping), all of which give rise to bearing errors.

Note - Where such bearing errors are known, AIP's will publish details: e.g. "Errors up to $5\frac{1}{2}^\circ$ may be experienced in sector 315° - 345° to 40 nm".

Designated Operational Coverage (DOC) –

- It is the range within which signal strength is guaranteed.
- Use of a VOR outside its DOC can lead to navigation errors.
- It is listed in AIP.
- Accuracy of $\pm 5^\circ$ on 95% of occasions (Day or Night).
- VOR DOC of 50/25 published in AIP means no interference within 50 NMS range up to 25,000'.

Factors affecting VOR Accuracy –

- a. **Beacon Alignment**- Radial 000 is not aligned with magnetic north.
- b. **Site error** is caused by uneven terrain and obstacles, in the vicinity of the transmitter. (Biggest Error)
- c. **Propagation error** - Having left the VOR site with $\pm 1^\circ$ accuracy, the transmissions are further affected by terrain over which they travel.
- d. **Airborne equipment errors** – Manufacturing Imperfections.
- e. **Aggregate Errors** – Errors due to propagation and airborne equipment errors combine to form aggregate errors. They can be calculated by pythagoras theorem. For example if propagation error is 3° and airborne equipment error is 4° , the aggregate error is 5° .
- f. **Pilotage error** - As an aircraft approaches the VOR the radials get closer together and difficulty can occur in maintaining or holding the desired radial.

Display –

1. Course Deviation Indicator – It shows the pilot where his selected track lies, either to his left or his right. It has 2 components –

- a. **Omni Bearing Selector (OBS)** –
 - It is used by the pilot to select a radial they wish to fly and maintain it.
 - This can also be used to find the radial an aircraft is flying.
 - If the pilot centralizes the L/R needle, the CDI will read the radial.
 - The full scale deflection is 10° , with 4 or 5 dots each indicating 2.5° or 2° each.
- b. **TO/FROM Indicator** – When the desired radial is selected on OBS, the TO/FROM Indicator will display TO if the selected radial is close to aircraft position and FROM if the selected radial is close to reciprocal of aircraft position.

Note – If the heading and selected radial are in the opposite direction, CDI will reverse sense the signal.

2. Radio Magnetic Indicator –

- It accepts relative bearing from ADF and phase difference from VOR & displays both.
- The head of the needle shows the QDM of the beacon, and the tail shows the radial on which the aircraft is positioned.

Warning Flag – Usually a red OFF flag appears in case of:-

- a. Failure of aircraft's equipment.
- b. Failure of ground station equipment.
- c. No power or low power supplied to aircraft.

The Cone of Ambiguity/ Cone of Confusion –

- As the VOR is approached, the radials converge and the VOR needle becomes more sensitive.
- Overhead the VOR the needle oscillates rapidly
- This is because there is no transmission vertically overhead in the shape of a cone. But some weak signals might be there.
- The indications of aircraft in cone of confusion are:-
 - ✓ The 'OFF' flag may appear momentarily.
 - ✓ The 'TO/FROM' needle flicker rapidly.
 - ✓ CDI needle oscillate left and right.

VOR SUMMARY

Characteristics	Magnetic bearings, valid day and night
Frequency	108 to 117.95 MHz; 160 channels
Uses	Airways; Airfield let-downs; Holding points; En-route navigation
Principle of Op	Phase comparison of two 30 Hz signals
Identification	3 letter aural Morse or Voice every 10 s, continuous tone for VOT (also ATIS using AM on voice)
Monitoring	Automatic site monitor +/- 1° Ident suppressed when standby transmitter initially switched on
Types	CVOR - reference signal is FM; variphase signal is AM - Limacon polar diagram rotating clockwise DVOR - more accurate than CVOR due to less site error - reference signal is AM; variphase signal is FM - simulated anticlockwise rotation of aerial TVOR - low power Tx at airfields VOT - Test VOR giving 180 radial - aircraft should have < +/- 4° error
Operational range	Transmitter power Line of sight DOC valid day and night
Accuracy affected by	Site error (less with DVOR) Propagation error Scalloping (bending due to reflections from terrain) Airborne equipment error (+/- 3°)
Cone of confusion	OFF flag may appear; TO / FROM display and bearings fluctuate
Airborne equip	Aerial, Receiver, Display (CDI / RMI) CDI: 2° per dot; max 10°; relationship between indication and aircraft position RMI: arrowhead gives QDM; tail gives QDR; Use magnetic variation at station
In Flight procedures	Radial interceptions; Track keeping; Station passage

Instrument Landing System (ILS)

ILS is a pilot interpreted runway approach aid. The runway served by ILS is called precision approach runway. ILS has 3 main components –

- i. **Localiser** – Used for approach guidance in horizontal plane along the extended centerline.
- ii. **Glideslope** - Used for approach guidance in vertical plane.
- iii. **Marker Beacons** – They are provided to enable the pilot to cross check the aircraft's height against ranges and timing to the runway threshold.

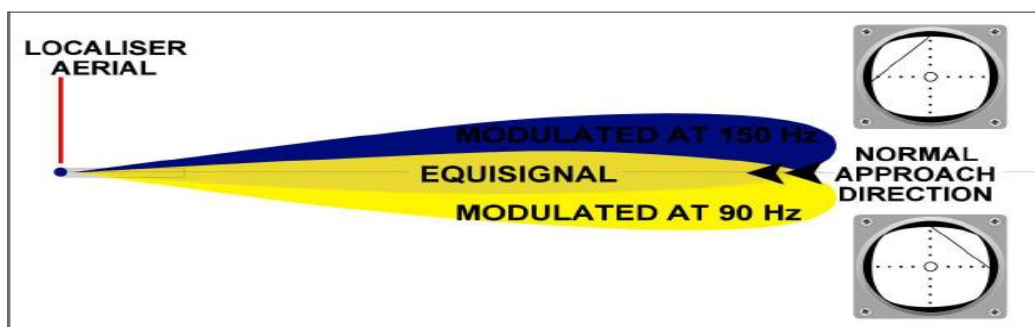
Localiser –

Location:

The Localiser (LLZ) is located about 300m from the up- wind end of the runway.

Principle:

- It produces two overlapping lobes along the runway approach direction in same frequency but modulated differently.
- The right hand lobe (the blue sector) has a 150Hz modulation.
- The left hand lobe (the yellow sector) has a 90Hz modulation.
- An aircraft approaching the runway centre line from the right will receive more of the 150 Hz signal than the 90 Hz modulation and vice-versa.
- This will result in vertical needle indicating a left turn.
- The difference in depth of modulation (DDM) will be zero along the runway centerline.



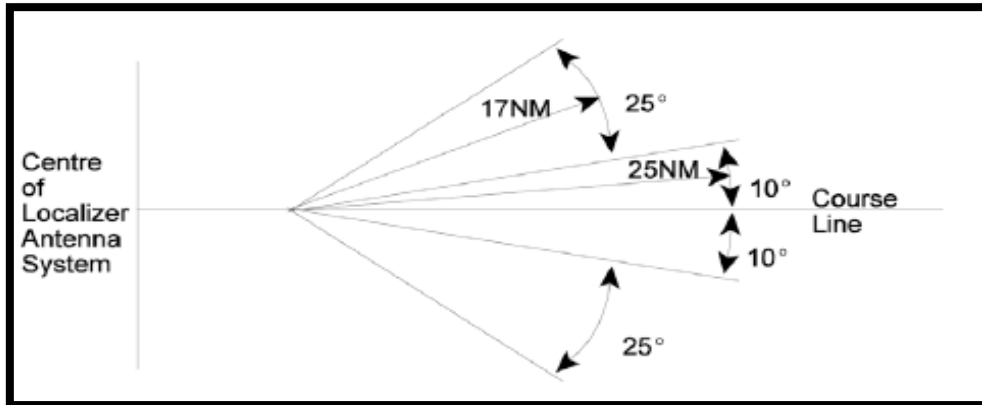
Frequency:

- The Localiser operates in the VHF band between 108 and 111.975MHz to provide 40 channels.
- This part of the frequency band is shared with VOR: the ILS frequencies have an odd number in the first decimal digit. e.g. 108.1 108.15; 108.3 108.35; 108.5 108.55 - 111.95 Mhz.

Coverage:

The localiser coverage sector extends from the transmitter to distances of:

- 25 nm within plus or minus 10° from the centre-line.
- 17 nm between 10° and 35° from the centre-line.
- 10 nm outside $\pm 35^\circ$ if coverage is provided.

**Glideslope****Location:**

The glideslope transmitter is located to one side of the runway approximately 150m from the runway edge, 300m upwind of the threshold.

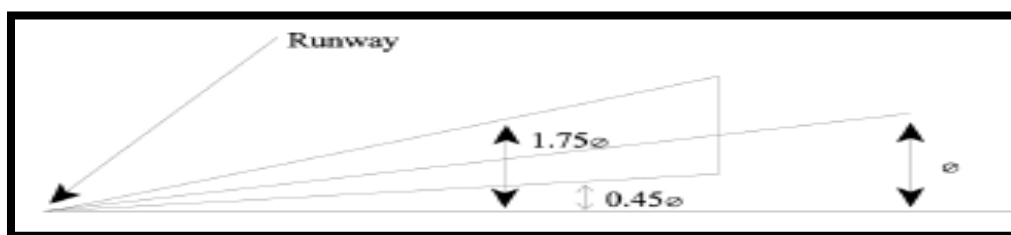
Principle:

- The same principle is used as for the localiser, except the lobes are in the vertical plane.
- The upper lobe (large lobe) has a 90Hz modulation, and the bottom lobe (small lobe) has a 150 Hz modulation.
- The DDM is zero at selected glide path usually 3° .

Frequency: The Glideslope operates in the UHF band between 329.15 and 335MHz to provide 40 complementary channels.

Coverage:

- In azimuth, coverage extends from 10 nm in sectors of 8° on each side of the centre-line.
- The vertical coverage is provided from 0.45ϕ up to 1.75ϕ above the horizontal where ϕ is the promulgated glide path angle.



Fan Markers

- 2 or sometimes 3 marker beacons are installed along the extended centerline to give range indications.
- These include the outer marker (OM), the middle marker (MM) and possibly an inner marker (IM).
- Marker beacons transmit at 75 MHz in the VHF band.
- They radiate a fan pattern towards a height of approximately 3000ft.

	Cockpit Light	Ident	Modulating Frequency	Pitch	Touchdown Range
OM	Blue	2 dashes/sec	400 Hz	Low	6.5 – 11.1 km (3.5 – 6 nm)
MM	Amber	Alternate dots and dashes 3/sec	1300 Hz	Medium	1 km (3500ft)
IM	White	6 dots/sec	3000 Hz	High	75 – 450 m (250 – 1500 ft)

Note –

- A DME that is frequency paired with an ILS supplements or replaces the range information from markers/locators.
- The DME ranges are zero referenced to the ILS runway threshold.
- The DME is protected only within the ILS localiser service area up to 25,000 feet.

MONITORING OF ILS TRANSMISSIONS

The monitoring equipment will act when:

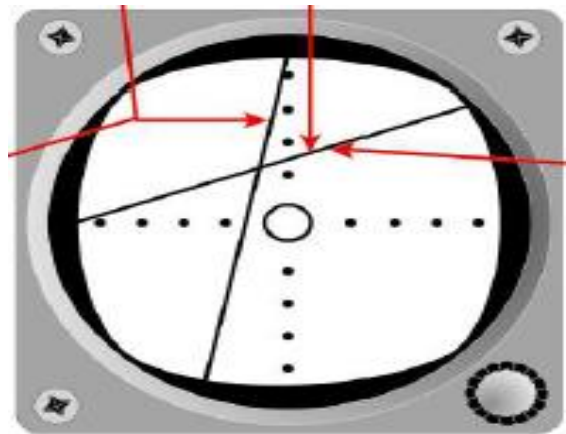
- The localiser shifts from the runway centre-line by more than 35'
- The Glide path angle changes more than $0.075 \times$ basic glide path angle;
- There is a power reduction in output of more than 50% from any transmitter.

The monitoring unit will provide warning to a control point and cause any of the following to occur:

- Stop all transmissions.
- Remove Ident.
- May permit operation to the lower category.

Localiser and Glidepath Indication –

- The Localiser deviation scale gives indications upto 2.5° on either side of the centerline. So on a 5 dot indicator each dot is 0.5° and on a 4 dot indicator it is 0.625° .
- The Glidepath deviation scale gives indications upto 0.7° above or below the glidepath. So on a 5 dot indicator each dot is 0.14° . Note that the maximum safe deviation below the glideslope is half full-scale deflection i.e. 2.5 dots fly up.



Type of Emission – A8W for Localiser, Glideslope and Marker Beacons.

ILS Identification –

- As the localiser and glidepath frequencies are paired both need not identify themselves separately.
- The selection of the localiser VHF frequency automatically energises the glide path receiver circuits.
- The Ident on the localiser is AM by horizontally polarized 1020 Hz. Transmission is a 2 or 3 letter morse code.

Limitations of ILS –

The Instrument Landing System has several limitations in that indications can be affected by:

- Beam bends caused by atmospheric conditions
- Scalping caused by reflections.
- Beam noise generated by the transmitter.
- High cost of installation.

False Glideslope –

- There could be more than one loci of points having DDM zero other than the correct ILS glidepath.
- The twin lobes are repeated due to:
 - Metallic structures situated at the transmission point, and ground reflections.
 - The propagation characteristics of the radiated lobe of glidepath is repeated several times above the true glidepath.
- The first false glideslope occurs at approximately twice the glide path angle, 6° above ground for a standard 3° glide path.
- False glideslopes always occur above the true glideslope and pilots should be aware of their presence.
- Normal flying practice is to establish on the localiser and intercept the glideslope from below to avoid false glideslope.

ILS Reference Datum Point -

The ILS reference datum point is a point at a specified height (around 50 feet) located vertically above the intersection of the runway centre-line and threshold, through which the downward extended portion of the ILS glidepath extends.

Visual Glide Path Indicators -

- The approach light systems such as PAPIs give a visual indication of the glide path to the runway that would be the same as that for the ILS.
- Its purpose is to give similar indications of glide path to the pilot from both systems during the final phase of the approach.
- 2 red and 2 white will mean the aircraft is on correct glidepath.

ILS CALCULATIONS –

- ❖ $\text{Height} = \text{Glidepath Angle} \times \text{Range} \times 101$
- ❖ $\text{ROD} = \text{Glidepath Angle} \times \text{Ground speed (kts)} \times 100/60$

ILS SUMMARY –

Components and frequencies:

Localiser	VHF - 108 to 111.975 MHz (40 channels). Aerial at upwind end.
Glide path	UHF - frequency paired. Aerial abeam touchdown.
Markers	VHF - 75 MHz. Fan shaped vertical radiation. OM, MM and IM.
Back beam	From localiser. Non-precision approach.
Locator	Low power NDB at OM.
DME	Freq paired. Possibly in place of markers. Zero-referenced to threshold.
Ident	2 or 3 letters, 7 groups/min. Suppressed when ILS u/s.
Markers	OM: blue 2 dashes/s 400 Hz 6.5 - 11.1 km MM: amber 3 characters per second, alternate dots and dashes 1300 Hz 1050m IM: white 6 dots/s 3000 Hz 75 - 450m
Ground monitoring	Localiser within 35' (Cat I) at ref datum. GP within 0.075 ϕ . Power within 50% Otherwise: Cease radiation, remove ident or lower category
ILS Coverage	LLZ: 25nm $\pm$ 10°, 17nm $\pm$ 35° GP: 10nm $\pm$ 8°, 0.45 to 1.75 x glidepath angle.

Principle of Operation:

Localiser	LH lobe - 90 Hz, RH lobe -150 Hz; DoM increases away from c/l DDM is zero on c/l
Glide path	Upper lobe - 90 Hz, lower lobe - 150 Hz. DoM increases away from ϕ DDM is zero on ϕ
False GP	At multiples of glidepath angle. Be aware.
Ref datum	Height of GP over threshold
Indicators	CDI: 0.5°/dot; max 2.5° . Reverse indication on back course HSI: set course selector to front QDM for correct indications GP: 0.14°/dot; max 0.7° . max safe dev - 2.5 dots fly up (0.35°)
Errors	beam bends, scalloping, beam noise; restricted vehicle movements during low vis ops; check failure flags; monitor ident

Radar Principles

Radar stands for **RA**dio **D**etection **A**nd **R**anging. It uses pulses and continuous wave techniques for its operation.

Types of Pulsed Radars

- i. **A Primary Radar** uses pulses of radio energy reflected from a target i.e. it uses one frequency throughout.
- ii. **A Secondary Radar** transmits pulses on one frequency, but receives on a different frequency i.e. the object transmits its own energy. It is a system utilising an interrogator and transponder; the transponder can be located in the aircraft or on the ground.

Applications of Radar -

1. **Air Traffic Control** uses radar to monitor aircraft, to vector the aircraft if necessary and provide information regarding weather e.g. storm clouds.
2. **Air/Ground navigational systems** use radar - Secondary Surveillance Radar provides ATC with information regarding an aircraft's callsign, altitude, speed, track history, destination and type of emergency when appropriate.
3. **Distance Measuring Equipment (DME)** provides a pilot with very accurate slant ranges from a ground based receiver/transmitter known as a transponder.
4. **Doppler Radar** is a self-contained airborne system, needing no ground based equipment, which provides a pilot with a continuous indication of the aircraft's drift and groundspeed.
5. **Airborne Weather Radar (AWR)** is used to:
 - a. Depict the range and bearing of clouds.
 - b. Indicate areas of the heaviest precipitation and associated turbulence.
 - c. Calculate the height of cloud
 - d. Ground map.

RADAR Frequencies –

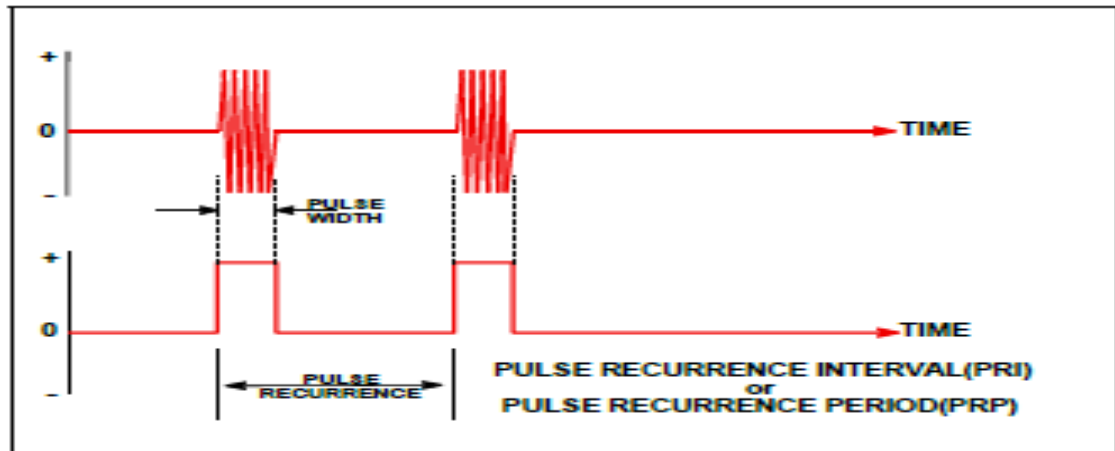
Radar systems are in the VHF and above frequency bands because:

- These frequencies are free from external noise/static and ionospheric scatter.
- The shorter wavelengths produce narrow, efficient beams for target discrimination and bearing measurement.
- Efficient reflection from an object depends upon its size in relation to the wavelength shorter wavelengths are reflected more efficiently.

Pulse Technique -

- Primary, secondary, and Doppler radar systems use the pulse technique which is the transmission of radio energy in very short bursts.
- Each burst of energy is in a pulse form of a pre-determined shape.

- The duration of the pulse is equal to the pulse length or width. Although a pulse is of short width (time) it can contain many cycles.



- Pulse Recurrence Interval (PRI)** is the time interval between two pulses. It is also known as the Pulse Recurrence Period (PRP)
- Pulse Recurrence Frequency (PRF)** is the number of pulses transmitted in one second (PPS). It is also known as the Pulse Recurrence Rate (PRR).
- Example. If the PRF is 250 PPS what is the PRI of the transmission?

$$\text{PRI} = 1 / 250 \text{ s}$$

$$\text{PRI} = 1,000,000 / 250 \mu\text{s} = 4000 \mu\text{s}$$

Theoretical Maximum Range –

- ❖ Maximum Range = $\frac{C}{2 \text{ PRF}}$
- ❖ Maximum Range = $\frac{\text{Pulse Length in seconds} \times C}{2}$

Examples

- We wish a radar to measure a range of up to 187 km. What should the PRF (PRR) be? (802 pps)
- What is the maximum PRR for a radar required to measure up to 200 nm? (405 pps)
- Maximum range for a radar is to be 170 km. What is the maximum PRR? (882pps)
- An AWR has a 400 pps PRR. Calculate the maximum range in nautical miles for this equipment. (203nm)

Factors affecting range of Primary Radar –

1. **Transmission power** - A radar signal attenuates with increasing distance from the transmitter. As the signal has to travel out and back the power/range relationship is:

$$\text{Range} = \sqrt[4]{\text{Power}}$$

The power needs to be increased by a factor of 16 to double the range.

2. **Aircraft height and the height of the radar head -**

Max. Theoretical range (nm) = $1.25 \times (\sqrt{\text{HTX}} + \sqrt{\text{HRX}})$

HTX = Height of radar station in feet amsl; HRX = height of target in feet amsl.

AIRPORT SURFACE MOVEMENT RADAR (ASMR)

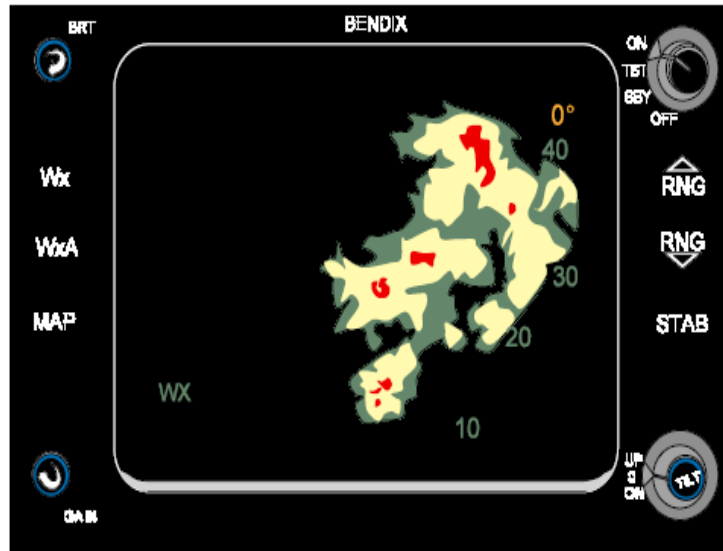
- This is also known as AIRFIELD SURFACE MOVEMENT INDICATOR (ASMI) and provides a very accurate radar display (in all weathers and conditions of visibility) of the aerodrome infrastructure, (taxi-ways, runways, aprons etc.), vehicular traffic and aircraft that are stationary, taxi-ing, landing or taking -off.
- ASMI radar is designed to provide a detailed, bright and flicker-free display of all aircraft and vehicles on runways and taxi-ways so that Air Traffic Control Officers can be certain that runways are clear of traffic before landings or take-offs, and to enable them to ensure the safe and orderly movement of traffic on taxi-ways.
- The very high definition required by these radars is achieved by designing a radar with:
 1. A very narrow beam in the order of 0.2° to 1° .
 2. A scanner rotation rate of 60rpm.
 3. A PRF in the order of 4,000 to 20,000pps.
 4. Pulse widths in the order of .03 μsec .
 5. Frequencies of 15 to 17 GHz (SHF), 2 to 1.76cm wavelengths.
 6. Ranges of 2.5 to 6nm in light precipitation.

Airborne Weather Radar

The Airborne weather radar (AWR) is used to provide pilots with information regarding weather ahead as well as navigation. The radar information can be displayed on a dedicated unit or shown (on modern aircraft) in combination with the aircraft route on the EFIS navigation display (ND).

AWR Functions - The main functions of an AWR are to:

- Detect the size of water droplets and hence deduce where the areas of turbulence are within the cloud.
- Determine the height of cloud tops by tilting the radar beam up or down.
- Map the terrain below the aircraft to provide navigational information and high ground avoidance.
- Provide a position fix (range and bearing) from a prominent feature.



PRINCIPLE OF OPERATION

Primary Radar

AWR is a primary radar and both of its functions, weather detection and ground mapping, use the echo principle to depict range and the searchlight principle to depict relative bearing of the targets.

Antenna

- The radar beam is produced by a suitable antenna in the nose of the aircraft.
- The antenna shape can be **parabolic or a flat plate** which produce both a **conical or pencil-shaped beam** as well as a **fan-shaped or cosecant square beam**.
- The type of radiation pattern will depend upon the use; the **pencil beam** is used for weather and **long range** (> 60 nm) mapping while the **fan-shaped** beam is used for **short range** mapping.
- It is usually necessary to tilt the antenna down when using the radar in the mapping mode.

Radar Frequency

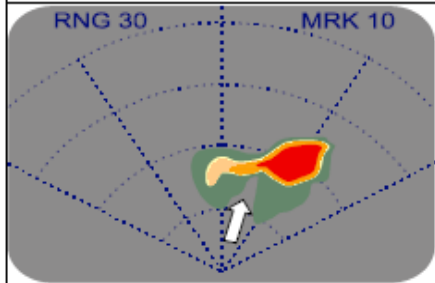
- The optimum radar frequency is one that has a wavelength comparable to the size of the large water droplets and wet hail which in turn are associated with severe turbulence; these droplets are about 3 cm across.
- The typical frequency adopted by most commercial systems is **9375 MHz, +/- 30 MHz** as it produces the **best returns from the large water droplets and wet hail** found in convective clouds.

In colour weather radar systems the weather targets are colour-coded according to the intensity of the rainfall as follows:

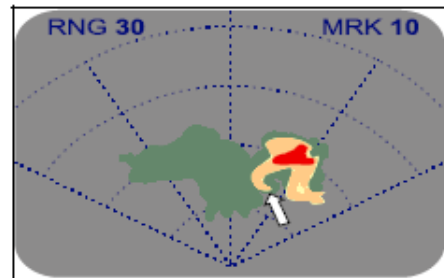
Colour	Pilot Interpretation	Amount of Rainfall
BLACK	Very light or no returns	Less than 0.7mm/hr.
GREEN	Light returns	0.7 - 4mm/hr.
YELLOW	Medium returns	4 - 12mm/hr.
RED	Strong returns	Greater than 12mm/hr.
MAGENTA	Turbulence	Due to rainfall intensity.

Note –

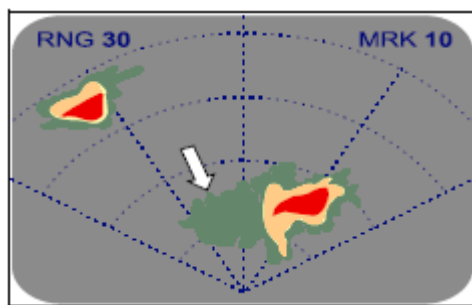
- The areas of greatest potential turbulence occur where the colour zones are closest together i.e. the steepest rainfall gradient.
- Also turbulence is associated with the following shapes: U-shapes, Fingers, Scalloped edges and Hooks. These are areas to avoid.



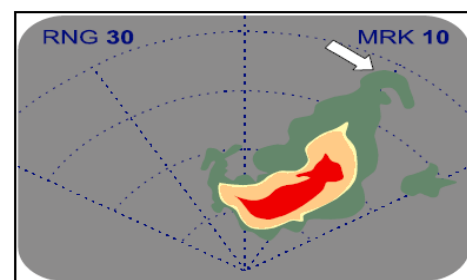
U-Shape



Finger

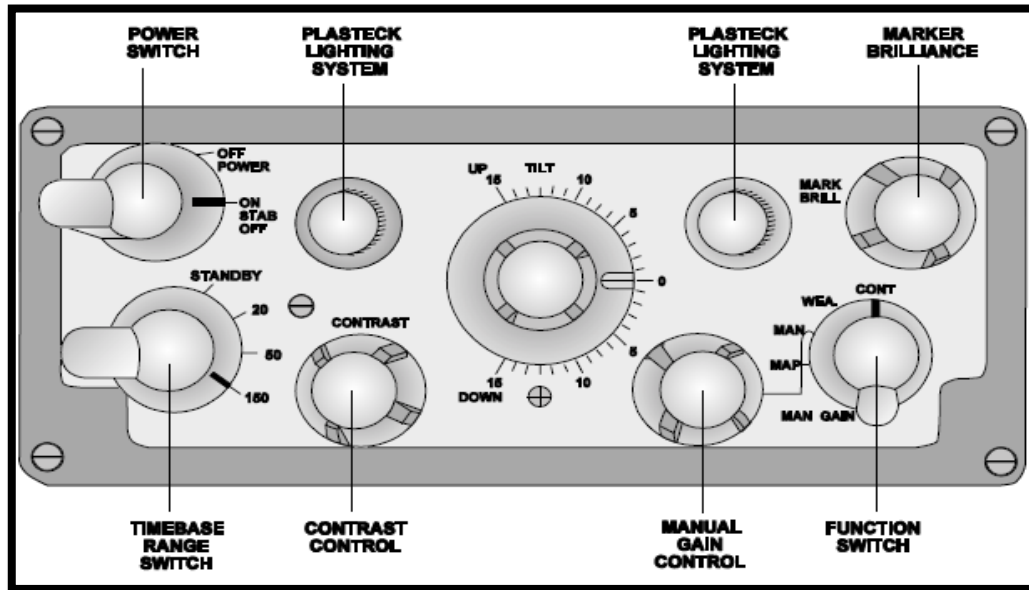


Scalloped Edge



Hook

Controls -



Power Switch

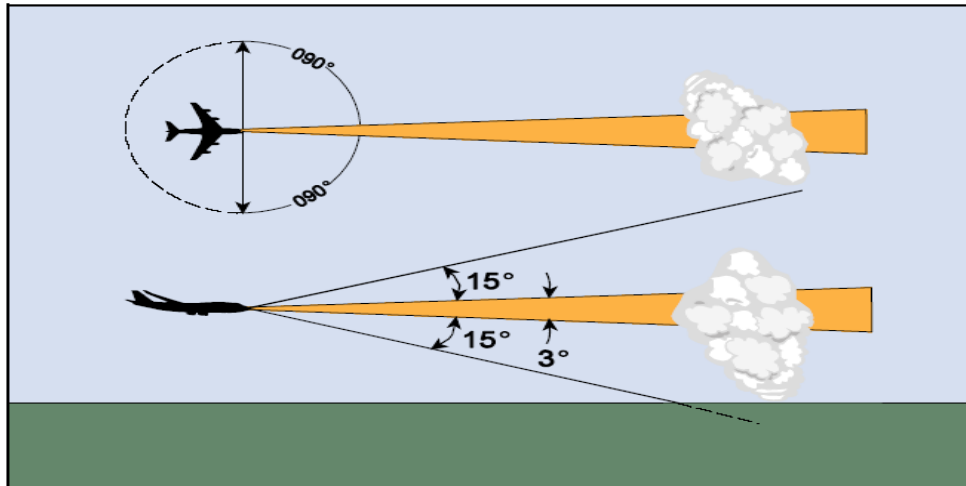
- In the ON position the system is energised and the aerial is automatically stabilised in PITCH and ROLL.
- Switching to the STAB OFF position will lock the scanner to the pitch and roll axes of the aircraft.

Range Switch

- Selection of a range position energises the transmitter.
- Whilst on the ground the STANDBY position must be maintained until it is certain that personnel and any reflecting objects, such as hangars, are not in the radar's transmitting sector.
- The radiation can damage health and the reflections from adjacent structures can damage the equipment.
- The maximum practical range for weather and for navigation is in the region of 150 nm.

Tilt Control

This control enables the radar beam to be tilted from the horizontal within 15° UP (+) and 15° DOWN (-).



Function Switch -

1. MAP Mode

- In the MAP position the radar produces a mapping beam.
- To obtain an even presentation of surface features, the transmitted power is progressively reduced as distance decreases so that the power directed to the closest object is minimum.
- This reduction in power with decreasing range is a function of the cosecant of the depression angle - hence the name cosecant beam; another description is "fan-shaped" beam.
- Signal amplification is adjustable via the adjacent MANUAL GAIN knob.
- Range is minimum (15nm) and maximum (60 to 70nm) depending upon the aircraft's height and type of terrain.

2. MAN Mode

- To map beyond 70nm the conical pencil beam should be used by selecting the MANUAL position.
- This is used for cloud detection and mapping between about 70 and 150nm.
- Selects the conical pencil shaped beam for long range mapping.
- MANUAL GAIN for signal amplification is operative with this selection.

3. WEA Mode

- This selects the conical pencil beam and is the usual position for observing cloud formations.
- MANUAL GAIN control is now INOPERATIVE.
- Instead a facility called Swept Gain, Sensitive Time Control or Automatic Gain Control (AGC) is automatically available.
- It operates up to about 25nm and ensures that the intensity (brilliance) of display of a particular cloud is independent of range.

- Thus a small cloud at 5nm does not give an increasingly stronger return than a larger and more dangerous cloud at 20nm; all clouds up to about 25nm are thus compared on equal terms.

4. CONT Mode

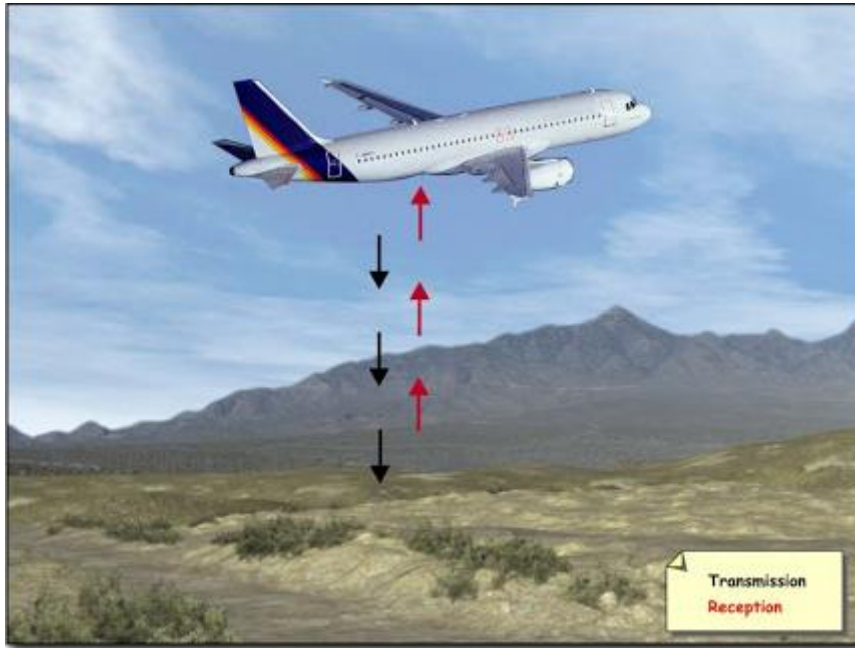
- It is a cloud formation presentation with CONT (CONTOUR) selected for a black and white display.
- The black holes indicate dangerous areas of concentrated rainfall and potential turbulence.
- It is used to indicate storm intensity, turbulence and areas to avoid.

Calculating Height of a cloud –

$$\text{Height} = \frac{\text{TILT} - \text{BEAMWIDTH}}{2} \times \text{Range} \times 101 \text{ feet}$$

Radio Altimeter

- The Radio Altimeter is a device designed to give indication of actual height above the ground.
- They are used at low levels, from ground to 2500ft.



- The instrument makes use of primary radio principles transmitting a **Frequency Modulated Continuous Wave (FMCW)** in an elliptical pattern vertically below the aircraft.
- The Radio Altimeter determines the time taken for a radio wave to travel from the aircraft to the ground directly beneath the aircraft and back again.
- During this time the transmitted frequency changes at a known rate from its start level to + 50 Mhz and back again to complete a “cycle”.
- Frequency used is 4200 MHz to 4400 MHz - SHF band.

Height Display

In analog the scale is logarithmic being expanded from zero to 500 feet and at a reduced non-linear scale from 500 to 2500 feet. Digital displays are used along with EFIS.

Mask - The height pointer disappears behind a mask:

- ✓ When altitude exceeds 2500 feet
- ✓ When there is any fault in the transmitted signal
- ✓ When the altimeter is switched off.

Failure Warning Flag

The flag appears when there is too much radio noise which will corrupt the returning signal, or if local reflections are received from the airframe itself, or in the event of a loss of power to the equipment.

Low Height Warning

- The warning light illuminates if the aircraft is flown below any pre- selected height set by the pilot.
- The pilot sets his desired height on this instrument by means of the moveable index (DH) control knob on the face of the device.
- This occurrence is also audibly marked by the sudden cessation of an alert tone which will sound with increasing loudness from approximately 100 feet above the decision height setting.

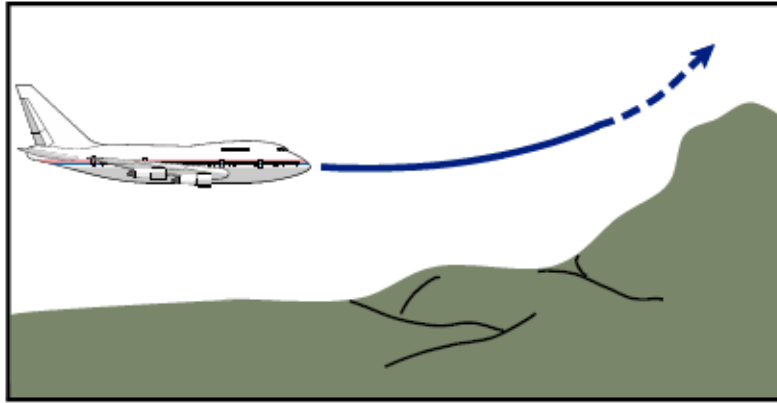
Range and Accuracy -

The instrument can be used between zero and 2500 feet above the surface with an overall expected accuracy of $\pm 3\%$ of indicated height or $\pm 1\text{ft}$ whichever is the greater.

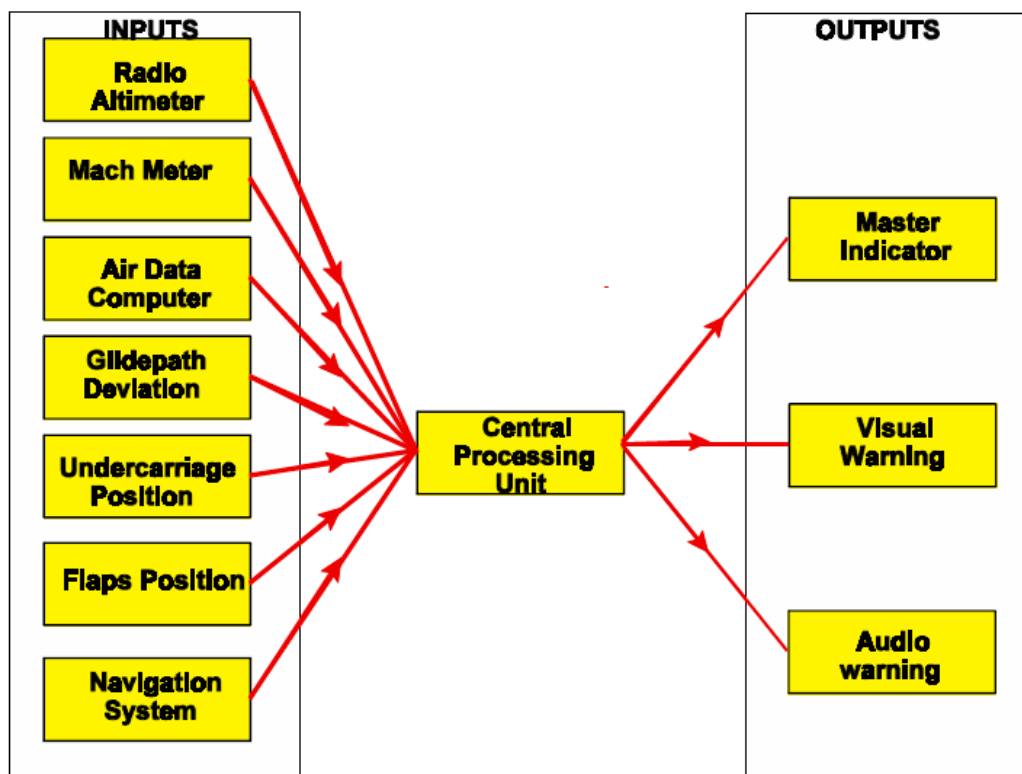
Altitude Alerting System –

- This is an altitude warning system which gives warning to the pilot of approaching selected altitude and also when deviating from the selected altitude.
- The alerts are visual and/or aural. It is similar to DH but is used in conjunction with pressure altimeter and takes inputs from the air data computer.
- The altitude selector knob is at the bottom right of the equipment.
- As the a/c approaches the selected altitude, 900ft before reaching it gets an audio warning. When 300ft from the selected altitude, the warning light goes out.
- Subsequently while flying at that altitude, if the a/c climbs or descends by more than 300ft, the light flashes and audio warning sounds.
- Once the a/c deviates by more than 900ft, all warning stop and the system resets itself.
- Such systems are mandatory for flying in RVSM airspace.

Ground Proximity Warning System **(GPWS)**



- GPWS is a warning device which alerts the pilot when the aircraft's proximity to the terrain poses a potential threat to its safety.
- It gives visual and audible warning signals
- The system operates between 50' and 2450' actual height above the surface and automatically selects the correct mode of operation.



GPWS Mode		Advanced Equipment	
		Alert (60 sec)	Warning (30 sec)
1. Excessive descent rate	50ft – 2450ft AGL	‘Sink Rate’	‘Whoop Whoop Pull Up’
2. Excessive terrain closure rate	50ft – 1800ft	‘Terrain Terrain’	‘Whoop Whoop Pull Up’
3. Altitude loss after take-off or go-around	50 – 700ft AGL	‘Don’t Sink’	-
4. Unsafe terrain clearance while not in the landing configuration	4a – Proximity to terrain - Gear not locked down (>500ft)	‘Too Low gear’	‘Too Low Terrain’
	4b- Proximity to terrain - Flaps not in a landing position (>200ft)	‘Too Low Flaps’	‘Too Low Terrain’
5. Descent below glideslope	Only below 1000ft AGL	‘Glideslope’	-
6. Descent below “minimums”	6a – Deviation from selected minimum.	‘Minimums’	-
	6b – Deviation from selected Bank Angle.	‘Bank angle’	
7. Windshear warning	-	-	‘Wind Shear’

MODE 1 - Excessive Descent Rate

- Mode 1 gives an aural alert of “SINK RATE” repeated each 1.5 seconds.
- The repeated warning of “WHOO, WHOO PULL UP” will occur until the rate of descent has been corrected.

MODE 2 – Excessive Terrain Closure Rate

- Aural alert - “TERRAIN, TERRAIN”
- Aural warning “WHOO, WHOO PULL UP”.

MODE 3 – Altitude loss after Take-Off or Go-Around

- Mode 3 provides an alert if a descent is made during initial climb or go-around.
- The aural alert is a voice message of “DON’T SINK”, repeated each 1.5 seconds until the flight condition is corrected.
- Mode 3 is effective between 50 and 700 feet radio altitude.
- Mode 3 does not arm during the descent until below 200 feet radio altitude.

MODE 4-**4A - Unsafe Terrain Clearance with Landing Gear not down**

- The terrain clearance mode with gear retracted, is armed after take-off upon climbing through 700 feet radio altitude.
- When this envelope is penetrated at less than 0.35 Mach, the aural alert "TOO LOW GEAR" is sounded.
- When the envelope is penetrated at more than 0.35 Mach, the aural alert "TOO LOW TERRAIN" is sounded and the upper boundary of the envelope is increased to 1000 feet radio altitude.

4B - Unsafe Terrain Clearance with flaps not in landing configuration.

- This mode provides an alert when the gear is down and the flaps are not in the landing position.
- If the envelope is penetrated at less than 0.28 Mach with the flaps not in the landing position, the aural alert of "TOO LOW FLAPS" is sounded.
- When the envelope is penetrated at more than 0.28 Mach, the aural alert of "TOO LOW TERRAIN" is sounded.

MODE 5 – Below Glideslope Deviation Alert

- This mode alerts the flight crew of a descent of more than 1.3 dots below an ILS glide slope.
- The mode is armed when a valid signal is being received by the captain's glide slope receiver and the radio altitude is 1000 feet or less.

MODE 6 –**6A - Below Selected Minimum Radio Altitude**

- Mode 6A provides an aural alert if a descent is made below the minimum decision altitude cursor in the captain's radio altimeter.
- This mode operates between 50 and 1000 feet of radio altitude.
- This alert is aural only and consists of "MINIMUMS, MINIMUMS" sounded once.

6B – Bank Angle Alert

- "BANK ANGLE" can be used to alert crews of excessive roll angles.
- The bank angle limit reduces with proximity to the ground due to the reduced wing tip clearance to prevent wing tip or engine damage during take-off and landing.

MODE 7 – Windshear Alerting

- Visual and aural windshear warnings are given when several parameters indicate the initial conditions of entering an area of windshear.
- The benefit from the system is derived from the fact that it allows the pilot to initiate the windshear go-around procedure earlier, giving the aircraft a greater probability of avoiding an accident.

Action to be taken on receiving a Warning.

The immediate response must be to level the wings and to initiate a maximum gradient climb which should be maintained until the aircraft attains the minimum safe altitude for that part of the route being operated.

Testing (BITE)

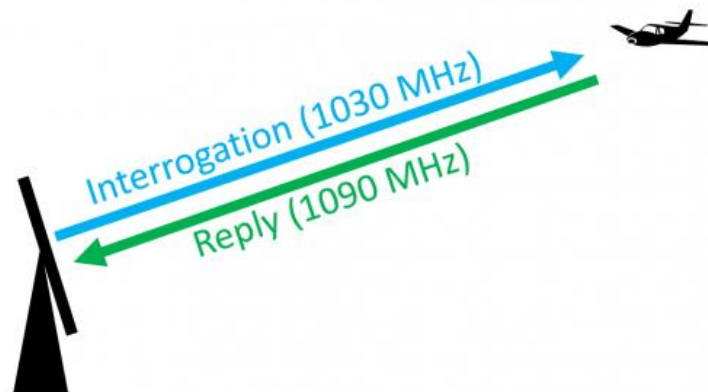
- The GPWS is provided with built-in test equipment (BITE) which allows all its functions and visual/audible warnings to be tested prior to flight.
- The test is initiated by the pilot pressing the test switch
- Testing is done only on ground.

Disadvantage – GPWS has no look ahead facility.

EGPWS or Terrain Awareness System (TAWS)

- Enhanced GPWS with the Terrain Threat Display.
- This terrain threat display and warning will be initiated in sufficient time to comfortably avoid any threat of flight into terrain.
- The terrain is shown in shades of green, yellow and red and the display indicates terrain not only below the aircraft but also ahead of its flight path.
- 1 minute before predicted impact the warning will issue a “Caution Terrain” message and the threat terrain will turn solid yellow.
- 30 seconds before collision warning of “Whoop Whoop Pull Up” will sound and the most threatening terrain will turn solid red.

Secondary Surveillance Radar (SSR)



Principle –

- The ground station is the interrogator which transmits a beam of rotating pulses.
- These beams are picked up by the aircraft transponder which decodes the aircraft signal and after a fixed delay replies on a different frequency.
- The aircraft identifies itself and gives its height.
- On the ground, range and bearing are displayed on the screen and other information such as A/c height, destination, point of leaving airspace and call-sign etc. are shown.

Frequencies- The ground station transmits/ interrogates on 1030MHz and receives on 1090MHz. The aircraft receives on 1030MHz and transmits/transponds on 1090MHz.

Modes - The signal from the interrogator is basically 2 pulses timed at specific intervals. The pulse intervals are called modes. The time intervals are –

Mode A – 8 μ sec

Mode B – 17 μ sec

Mode C - 21 μ sec

Mode D - 28 μ sec

Mode A and B are used for identification, mode C for automatic height information and mode D is experimental.

Functioning of pulses –

- Once the pilot decides to respond the transponder replies with a code obtained by inclusion or omission of up to 12 pulses.
- This train of 12 pulses is contained between 2 framing pulses, 20.3 μ sec apart. The number of codes that can be made by this train of 12 pulses is **4096**.
- These are numbered from **0000 to 7777** using all numbers except 8 and 9.
- Each of these pulses in the transponder are 0.45 μ sec wide.

- A further pulse called the **Special Position Identification (SPI)** pulse may be transmitted when the “Ident” button on the pilot’s transponder is pressed.
- This pulse is after the last framing pulse and will be automatically and continuously transmitted for about 20 seconds.
- It produces a distinctive display so that a controller can pick out a particular aircraft by asking the pilot to “**Squawk Ident**”.

MODE C -

- When the aircraft receives a Mode C interrogation the transponder will produce one of the 4096 codes.
- This code is decided as per the altitude the aircraft is flying.
- The code corresponds to its height is referenced to 1013.2mb, regardless of the pressure setting on the altimeter.
- The system provides Automatic Altitude Telemetry up to 128,000ft, with an output change every 100ft and provides the controller with the aircraft’s Flight Level or Altitude. For eg – If the aircraft deviates from its allocated level by a 100ft, the controller will automatically get updated about aircraft height.

Special Transponder Codes and their Meaning –

- 7700 – Aircraft in Distress (Mayday)
- 7600 - Radio Communication Failure.
- 7500 - Unlawful Interference (Hijack)
- 7000 - No code allotted by ATC (conspicuity).
- 7004 – Aircraft carrying out acrobatic maneuver.
- 2000 - To provide recognition of an aircraft which has not received any instructions from ATC/ Crossing FIR.
- 0000 – Aircraft Transponder Malfunction.

Advantages –

- Only one way transmission, requires much less power.
- Gives clutter free responses as it does not rely on returning reflected pulses.
- Positively identifies an aircraft’s primary response by displaying its code and call sign alongside.
- Indicates an aircraft’s track history, speed, altitude and destination.
- Can indicate on a controller’s screen that an aircraft has an emergency, has lost radio communications or is being hi-jacked.

Disadvantages -

- Garbling** - This is caused by overlapping replies from two or more transponders on nearly the same bearing from the ground station and within a distance of 1.7 nm from each other measured on a line from the antenna.
- Fruiting** - This is interference at one interrogator caused by replies from a transponder responding to interrogations from another. This can be avoided by

allotting neighboring radars different PRFs and disregarding a sequence of returns at the wrong PRFs.

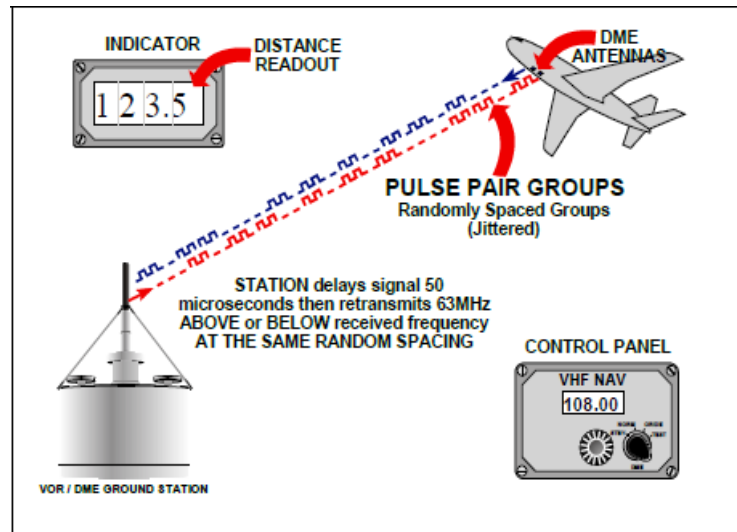
- c. **Over Interrogation** – If aircraft is being interrogated by more than one station, it may get saturated and not reply to any of them.

MODE S

- Mode S is being introduced in order to overcome the limitations of the present modes A and C.
- The system will have over 16,700,000 discrete codes available for allocation to individual aircraft on a permanent basis.
- The uplink messages will be specifically addressed to a particular aircraft.
- This will eliminate problems like garbling, fruiting and over interrogation as each aircraft will be monitored separately.
- This will be in 25ft increments and more data on an aircraft's present and intended performance will be available.
- Enhanced information in the form of bank angle, accelerations, heading, groundspeed will be passed automatically to the ATC.

Distance Measuring Equipment (DME)

- Distance Measuring Equipment (DME) is a secondary radar system which provides accurate slant range of the aircraft from the ground transmitter.
- Interrogator is on the aircraft and transponder is on the ground.
- DME (emission code P0N) operates between 962 and 1213MHz in the UHF band.



Principle of operation - Pulse Technique (Random PRF).

- The aircraft wishing to interrogate and find its range sends a series of pairs of pulses at **random time intervals** on the ground stations frequency.
- The ground station receives these pairs of pulses and after a fixed time delay of 50 μ sec sends a reply in the same pattern (PRF) that the pair of pulses arrived.
- The aircraft equipment knows the pattern (PRF) of the pair of pulses that it had transmitted and when it gets a reply of a similar pattern but on a frequency that is ± 63 MHz different from the interrogation frequency.
- The airborne system measures the time interval, electronically, between the start of the interrogation and the reception of the response from the transponder, adding 50 μ sec will give the slant range.

Twin Pulses -

The use by the DME system of twin pulses ensures that the receivers never accept matching randomised single pulses which could (possibly) emanate from, for example, other radars, ignition systems or lightning.

Range Search -

- To achieve a rapid lock-on during the range search, the DME interrogator transmits at 150 pulse pairs per second for 15000 pulse-pairs (100 seconds).
- If lock-on is not achieved, it will then reduce the rate to 60 pps and maintains this rate until there is a range lock-on.
- At lock-on the system operates at a random PRF of 27pps.

Note –

- 1) Groundspeed calculated will be most accurate when the aircraft is flying away from beacon and on route DME is used.
- 2) If DME signals are not received due to terrain the before going back to search mode it will shift to memory mode. i.e – It will keep calculating the distance using previous groundspeed for 20 seconds.

Self-Triggering –

- The transmissions on ground can be reflected by objects and obstacles in its vicinity.
- Transponder will consider them as reply from the aircraft and send out ranges of these reflections. This process is called self-triggering.
- It is prevented in DME by using different frequencies for interrogation and response.

Beacon Saturation –

- DME beacon gets saturated when more than 100 aircraft trigger it at the same time.
- The output of the ground beacon is a constant 2700 pps random whether or not is being triggered by aircraft or not.
- As the aircraft transmits at 27pps, this will equate to about 100 aircraft using the DME at the same time.
- So, if there are more than 100 aircraft, the equipment will reply to the stronger signals by reducing its gain.

Ident –

- A 3-letter callsign is transmitted every 30 seconds, usually in conjunction with an associated VOR.
- The DME identification is distinguished from the VOR identification by having a different tone (usually higher than the VOR).

VOR/DME Frequency Pairing –

To reduce the pilot's cockpit workload, VORs are frequency paired with a DME. The conditions for frequency pairing are-

- i. Both transmitters Co-located
- ii. The maximum distance between both transmitters is 30m/100' in TMAs.
- iii. The maximum distance between both transmitters is 600m/2000'

Note - VOR/DME-TACAN widely separated i.e. >6nm may or may not be paired and have totally different identifications.

Advantages of DME-

- No interference due weather as reflection of cloud will not take place due to difference of 63MHz in Tx and Rx frequency.
- Power required will be less as 2 way travel not required.
- Self-triggering does not take place

Range and Coverage -

The range of DME depends on –

- Aircraft height
- Transmitter height
- Line of sight

System Accuracy - Based on a 95% probability the system accuracy for DME should give a total system error not exceeding ± 0.25 nm $\pm 1.25\%$ of range.

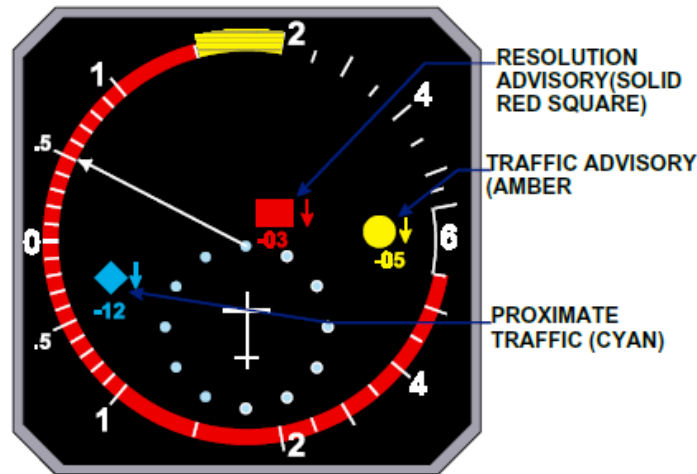
Range – Almost 200nms.

TACAN (TACtical Air Navigation) - It is a military nav-aid and uses DME to provide ranges. It cannot be used by civil aircrafts but its DME can be used by civil aircrafts.

DME SUMMARY

Frequency	UHF band; 962 to 1213 MHz; 1 MHz spacing; 252 channels +/- 63 MHz difference between transmitted and received frequencies Selection by paired VHF frequency (VOR or ILS). DME paired with ILS gives range zero referenced to ILS runway threshold
Uses	Groundspeed and time to / from station. DME arcs range and height checks during let-downs accurate ranges to threshold RNAV.
Principle of Operation	Aircraft interrogator and receiver: transmits pairs of pulses at random intervals, omni-directionally. Ground station transponder: re-transmits all pulses at +/- 63 MHz after a delay of 50 μ s.
Slant Range	Aircraft receiver identifies own pulses and determines range from time interval between transmitted and received pulses (-50 μ s).
Pulse characteristics	Twin pulse used to avoid interference Jittered pulses are used to identify own pulses Frequency change prevents aircraft locking on to reflections
Range Search	Pulse rate - initially 150 pps - reduced to 60 pps after 15,000 pps - further reduced to about 25 pps at lock-on
Beacon saturation	Occurs at 2,700 pps (approx 100 aircraft interrogating) receiver gain reduced to respond only to strong pulses
Station Ident	3 letter identifier; range info not available during ident period
Coverage	Line of sight range; reduced by intervening high ground and bank angle DOC gives protected range; echo protection circuit eliminates reflections
Accuracy	+/-0.25 nm +/-1.25% of range (+/-0.2 nm for precision systems) Slant range error significant when aircraft range < 3 x height; Groundspeed error increases as aircraft goes higher and closer to station.
VOR /DME Frequency Pairing	Associated - if co-located - within 100' in TMA or 2000' outside TMA - callsigns are the same; frequencies paired Not associated - if serving same location then callsign of DME third letter is 'Z' - frequencies paired Separated - if > 6 nm apart; callsigns different

Traffic Collision and Avoidance System (TCAS)



- **Principle** – Secondary Radar
- **Range** – 30nm
- **Frequency** – 1030 & 1090 MHz

TCAS I

- First generation collision avoidance system and simply warns the crew of other traffic in the vicinity of their aircraft.
- Normally gives about 40 seconds warning of a possible collision.
- It will detect and display range and approximate relative bearing.
- It encourages flight crew to look for the conflicting traffic by generating visual and aural warnings - TRAFFIC ADVISORIES (TAs).
- Pilot has to take avoiding action as deemed fit.

TCAS II

- TCAS II detects intruders in the TCAS aircraft's vicinity, assesses the collision risk and presents warnings to the crew in the form of TAs and Resolution Advisories (RAs) in Vertical plane. Eg.: "Climb" "Increase Climb" Descend" "Increase Descent" Monitor Vertical Speed"
- Can track upto 45 aircraft and display upto 30.
- Can coordinate RA's upto 3 conflicting aircraft.
- Gives RA 20-30 seconds before estimated collision.
- Will not give RA's below 1000 ft AGL and nor an aircraft will be given RA to descend below 1000ft.
- TCAS III and ACAS III are future systems and will give horizontal RA's also.

Visual Display –

- **Traffic Advisories (TAs)** denoted by a Solid Yellow Circle.
A voice – “Traffic Traffic”
- **Resolution Advisories (RAs)** denoted by a red square.
Appears only if actual collision threat.
- ◆ **Blue Diamond (Cyan)** will denote proximate traffic which has no threat.
Range <6nm and <1000ft.
- ◇ **White Diamond (hollow cyan diamond)** denote traffic with no threat. Range
> 6nm or >1220ft vertically.

Area Navigation Systems (RNAV)

An area navigation (RNAV) system is any system that allows the aircraft to be navigated to the required level of accuracy without the requirement to fly directly over ground based facilities.

OPERATION OF A SIMPLE 2D RNAV SYSTEM

- A simple RNAV system uses range and bearing information from VOR/DME stations to define position.
- The pilot defines waypoints along the route to be flown as range and bearing from suitably located VOR/DME.
- Then the equipment, using the VOR/DME bearing and range, computes the QDM and distance to the waypoint and presents the information to the pilot on a CDI or HSI as if the waypoint itself is a VOR/DME station, hence these waypoints are known as phantom stations.

Note – Phantom Stations are the waypoints defined with respect to radials and distance. Eg- AKELA, EDNOR etc.

There are two types of RNAV:

- i. **Basic RNAV (B-RNAV)** which is required to give a position accuracy to within 5 nm on 95% of occasions. It is now mandatory for all aircraft carrying 30 passengers or more to have B-RNAV.
- ii. **Precision RNAV (P-RNAV)** must be accurate to within 1.0 nm on 95% of occasions. P-RNAV routes are now being established in terminal airspace.

There are three levels of RNAV capability:

- 2D RNAV which relates to the capabilities in the horizontal plane only.
- 3D RNAV indicates guidance capability in horizontal and vertical plane.
- 4D RNAV indicates the addition to 3D RNAV of a timing function.

Advantages of RNAV -

Less Distance, Less Time, Direct Route, Less Fuel, Less Cost.

Note -

- a) In case the aircraft is outside coverage RNAV shift to DR mode (Dead Reckoning Mode) i.e – Last Speed, Last Position.
- b) Cross- Track Distance is displayed on ND with the track keeping accuracy of 5nm in En-route mode and 0.5nm in approach mode.

Electronic Flight Information System

(EFIS)

The Electronic Flight Instrument System presents attitude and navigation information to the pilot on two electronic display units in a format that is easier to read and less likely to be misinterpreted than some older mechanical instruments.

As in the case of a multi crew conventional flight director system, a complete EFIS installation is made up of left (Captain), and right (First Officer), systems.

Each system in turn is comprised of:

- ✓ Electronic Attitude Director Indicator (EADI) or Primary Flight Display (PFD)
- ✓ Electronic Horizontal Situation Indicator (EHSI) or Navigation Display (ND)
- ✓ Control Panel
- ✓ Symbol Generator (SG)
- ✓ Remote Light Sensor Unit

The Colour Display System

1. GREEN - engaged flight mode displays, selected data, dynamic conditions.
2. White - present status situation, scales, armed flight mode displays
3. MAGENTA - command information, selected heading, symbols, fly-to condition
4. CYAN - (blue) non-active and background information, sky.
5. RED – warning
6. AMBER - cautionary information, faults, flags
7. BLACK - blank areas, display “off”
8. Brown – Earth
9. Yellow – Abnormal Sources

Modes of EFIS –

Full VOR –



Expanded VOR –



Full ILS



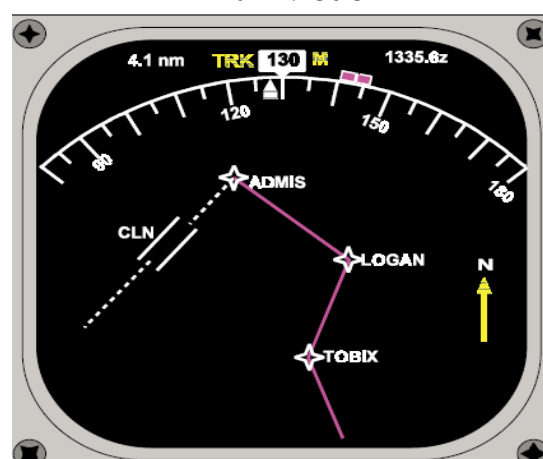
Expanded ILS



Map Mode



Plan Mode



Global Positioning System (GPS)

The NAVSTAR Global Positioning System (GPS) operated by the USA.

The Global Orbiting Navigation Satellite System (GLONASS) operated by Russia.

Global Navigation Satellite System (GNSS) operated by Europe.

- GPS is a satellite based system which calculates a 3D position in space with minimum 4 satellites giving you range.
- In case only 3 satellites are available then only latitude and longitude can be derived and height can be manually fed.
- The system uses 24 satellites in 6 orbits. In that, 21 are active and 3 are spare.
- Transmission is in UHF (L Band).

Principle of operation – Position fixed by range position line.

- The GPS is a satellite based system and uses principle similar to DME.
- The satellite sends their position and time signal.
- Receivers at a/c note the time difference between transmissions and receptions.
- Distance measured with time difference and thus range position lines are obtained.

The GPS Segments – GPS comprises three segments:

- i. The Space Segment
- ii. The Control Segment
- iii. The User Segment

The Space Segment

- In GPS the space segment comprises of 24 SVs in 6 orbital planes.
- 21 SVs are operational and the other 3 are in-space spares.
- The orbits have an average height of 10,898nm (20,180km) and have an orbital period of 12 hours.
- The orbital planes have an inclination of 55° and are equally spaced around the Equator.
- The SVs broadcast pseudo-random noise (PRN) codes of one millisecond duration on two frequencies in the UHF band and a NAV and SYSTEM data message. Each SV has its own unique code.
- L1 Frequency: 1,575.42 MHz transmits the coarse acquisition (C/A) code. This provides standard positioning service (SPS) and can be used by all users.
- L2 Frequency: 1227.6 MHz transmitting the Precise or P code. The second frequency is used to provide precise positioning service (PPS).
- The PRN codes provide SV identification and a timing function for the receiver to measure SV range.
- But the ranges are called Pseudo Ranges and are not accurate.

- This is due to the fact that the satellite uses an accurate atomic clock while the Rx uses an inaccurate quartz clock. So the ranges have to be made accurate and Time bias error needs to be removed.
- To remove this error the receiver takes fix from 3 satellites at a time then the set of next 3 and then the set of next 3. With this the receiver makes changes to its clock and removes the time bias which gives accurate position.

The Control Segment –

- The monitoring stations check the SVs' internally computed position and clock time at least once every 12 hours.
- The SV orbits are affected by the gravitational influences of the sun, moon and planets and are also affected by solar radiation, so errors between the computed position and the actual position occur.
- This segment detects malfunctioning of satellites.
- Accuracy is 95% times accurate.

The User Segment –

- GPS receivers use the space segment to determine position on and close to the surface of the Earth. These receivers may be stand-alone or be part of integrated systems.
- When Rx is switched on, it searches for signals on the correct frequency.
- Receiver which has no knowledge of its own position must download full set of almanac satellites. (12.5 min for downloading)
- After this the receiver searches for those satellites whose signal it can receive.
- Then it generates own pseudo-random codes and searches for satellite with similar code.
- When both code matches the receiver locks on to the satellite and now looks for next satellite and so on.

Failure Detection and Removal system

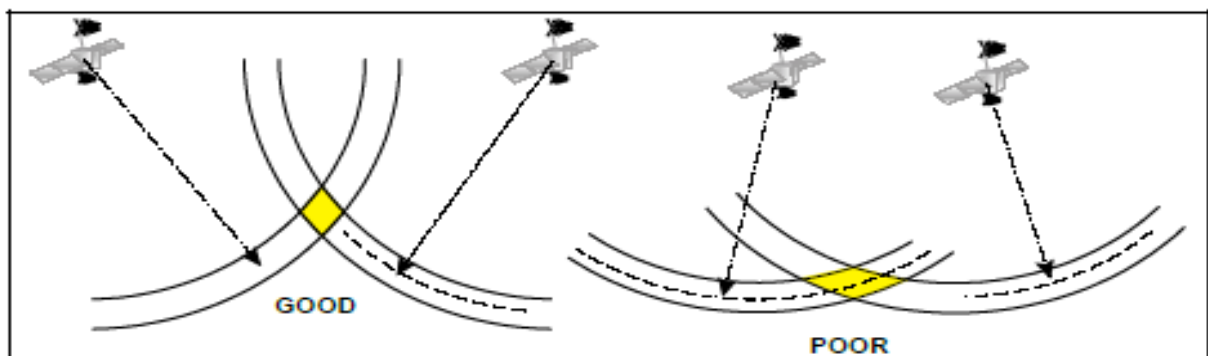
RAIM (Remote Autonomous Integrity Monitoring) –

- It requires 5 SVs to detect if any of the data from any of the SVs is in error and a minimum of 6 SVs so that if a SV giving wrong info can be excluded. This capability is called RAIM. $RAIM + FDE = 6 \text{ SVs}$
- If 6 satellites are not received a warning light will glow informing no RAIM.
- RAIM thus provides a warning system.

GPS ERRORS -

- Ephemeris Errors** – They are caused by the gravitational effects of the sun, moon, planets and solar radiation.
- Time Bias error** – 12.5 minutes time for download.
- Satellite Clock Error** – Errors in the atomic clock of the satellites. Maximum error 1.5 m.

- iv. **Ionospheric propagation error** - The interaction of the radio energy with the ionized particles in the ionosphere causes the radio energy to be slowed down as it traverses the ionosphere, this is known as the ionospheric delay.
- v. **Receiver noise error** - All radio receivers generate internal noise, which in the case of GNSS receivers can cause errors in measurement of the time difference. Maximum 0.3m.
- vi. **Multi-path reception** - Reflections from the ground and parts of the aircraft result in multipath reception. This can be minimised by careful siting of the aerial and by internal processing techniques. Maximum 0.6 m.
- vii. **Geometric dilution of precision (GDOP)** - This is the GPS term for a poor cut between position lines. GDOP occurs when the satellites are (relatively) close to each other.



- viii. **Selective availability (SA)** - USA deliberately degraded the accuracy of the fixing on the C/A code (i.e. for civilian users).

SYSTEM ACCURACY

The ICAO specification requires an accuracy (95%) of the SPS to be:

- Vertical: ± 13 m
- Horizontal: ± 22 m
- Time: 40 nanoseconds (10^{-9})

Differential GPS (DGPS) -

- DGPS is a means of improving the accuracy of GPS by monitoring the integrity of the SV data and warning the user of any errors which occur.
- DGPS systems will provide warning of failure in the SV data and prevent or minimize the effect of such errors, or provide failure warning and improve the accuracy of the deduced position.
- DGPS can remove selective availability error and group delay. But it cannot remove GDOP error.
- Major advantage is that it can provide external integrity monitoring and can supplement RAIM.

There are two kinds of DGPS currently in use: -

- i. **WAAS** (Wide area augmentation system) – Used for giving accurate enroute navigation information.
- ii. **LAAS** (Local area augmentation system) - This system is used in aviation to provide accurate navigation information for approach and landing.

The GPS-aided GEO augmented navigation (GAGAN) is an implementation of a regional satellite-based augmentation system (SBAS) by the Government of India. It is a system to improve the accuracy of a GNSS receiver by providing reference signals.

Flight Management System

Flight management systems are designed to improve navigation, aid fuel efficiency and to reduce crew workload.

- i. **LNAV** - Computers are used to fly aircraft along complex routes using Lateral Guidance.
- ii. **VNAV** - Vertical Guidance enables the system to calculate optimum cruise altitudes.

FMS has 2 database –

- a. Navigation Database - Airports, SIDs, STARs, Nav-aids, Terrain Data, Elevation.
- b. Performance Database – Climb Performance, Thrust Limits, Fuel Flow, Optimum altitudes.

Note - FMS Database is read only database to pilot and is valid for 28 days.

Sequence of pages in FMC is –

1. Ident
2. POS INIT
3. ROUTE
4. PERF INIT

Cost Index -

- It is determined by dividing aeroplane operating cost by fuel cost.
- It is the input code telling the FMS the required balance between optimizing fuel cost and sector time.
- A cost index of Zero results in maximum fuel saving but more time burn while cost index of 200 results in maximum time saving but more fuel burn.
- It is decided by the operator what to save.

Note –

- i. Kalman Filtering uses long term accuracy of GPS, short term accuracy of IRS.
- ii. Position displayed on GPS by FMC is the average of RNAV, GPS and self-contained Nav aid IRS.
- iii. FMC is most inaccurate at the Top of Descent (TOD).

Radio Aids Questions

1. **A radio wave is:**
 - a. an energy wave comprising an electrical field in same plane as a magnetic field
 - b. an electrical field alternating with a magnetic field
 - c. an energy wave where there is an electrical field perpendicular to a magnetic field
2. **The speed of radio waves is:**
 - a. 300 km per second
 - b. 300 million metres per second
 - c. 162 nm per second
3. **The plane of polarisation of an electromagnetic wave is:**
 - a. the plane of the magnetic field
 - b. the plane of the electrical field
 - c. the plane of the electrical or magnetic field dependent on the plane of the aerial
4. **If the wavelength of a radio wave is 3.75 metres, the frequency is:**
 - a. 80 kHz
 - b. 8 MHz
 - c. 80 MHz
5. **The wavelength corresponding to a frequency of 125 MHz is:**
 - a. 2.4 m
 - b. 24 m
 - c. 24 cm
6. **The frequency which corresponds to a wavelength of 6.98 cm is:**
 - a. 4298 GHz
 - b. 4.298 GHz
 - c. 429.8 GHz
7. **The frequency band containing the frequency corresponding to 29.1 cm is:**
 - a. UHF
 - b. VHF
 - c. SHF
8. **The wavelength corresponding to a frequency of 15 625 MHz is:**
 - a. 1.92 m
 - b. 19.2 m
 - c. 1.92 cm
9. **Which frequency band is a wavelength of 1200 m?**
 - a. UHF
 - b. LF
 - c. HF

Answers

1	2	3	4	5	6	7	8	9
C	B	B	C	A	B	A	C	B

Radio Propagation Theory Questions

1. **The process which causes the reduction in signal strength as range from a transmitter increases is known as:**
 - a. absorption
 - b. diffraction
 - c. attenuation
 - d. ionization
2. **Which of the following will give the greatest surface wave range?**
 - a. 243 MHz
 - b. 500 kHz
 - c. 2182 khz
 - d. 15 MHz
3. **It is intended to increase the range of a VHF transmitter from 50 nm to 100 nm. This will be achieved by increasing the power output by a factor of:**
 - a. 2
 - b. 8
 - c. 16
 - d. 4
4. **A 300 kHz transmitter has an output of 1600 watts, the effective range over the sea will be:**
 - a. 52 nm
 - b. 80 nm
 - c. 35 nm
 - d. 120 nm
5. **The maximum range an aircraft at 2500 ft can communicate with a VHF station at 196 ft is:**
 - a. 80 nm
 - b. 64 nm
 - c. 52 nm
 - d. 65 nm
6. **What is the minimum height for an aircraft at a range of 200 nm to be detected by a radar at 1600 ft amsl?**
 - a. 25,500 ft
 - b. 15,000 ft
 - c. 40,000 ft
 - d. 57,500 ft
7. **The average height of the E-layer is and the maximum range for skywave will be**

- a. 60 km, 1350 nm
- b. 125 km, 2200 km
- c. 225 km, 2200 km
- d. 125 km, 1350 nm

8. Concerning HF communications, which of the following is correct?

- a. The frequency required in low latitudes is less than the frequency required in high latitudes.
- b. At night a higher frequency is required than by day.
- c. The frequency required is dependent on time of day but not the season.
- d. The frequency required for short ranges will be less than the frequency required for long ranges.

Answers

1	2	3	4	5	6	7	8
C	B	D	D	A	B	D	D

Radio Aids DGCA Questions

- 1. Radio waves travel through the atmosphere _____ ?**
 - a. At a speed which varies in direct relation to power of the transmission
 - b. At a speed which varies directly as the frequency
 - c. At a constant speed

- 2. The period in which the potential of a radio wave rises from zero to a maximum positive and back to zero and to maximum negative and back to zero is known as:**
 - a. The frequency
 - b. A cycle
 - c. The wave length

- 3. The skip distance**
 - a. The difference between the aural null and maximum reflection
 - b. The distance between 1 and reflected skywave initially strikes the earth.
 - c. The distances between the point where the ground wave fades out, and reflected sky wave strike the earth.

- 4. An advantage of VHF compared to low frequency radio is**
 - a. Speed reception as the waves travel faster
 - b. Reduction in static interference
 - c. Much greater range is possible

5. **Advantage of VHF over MF and HF is that**
 - a. Static effect is reduced
 - b. Static effect is more
 - c. No difference
6. **In a microsecond of a radio wave will travel approximately:**
 - a. 2000'
 - b. 1000'
 - c. 500
7. **If the frequency is 600 khz the wavelength is**
 - a. 500 cms
 - b. 50 mts
 - c. 500 mts
8. **The best frequency band to use in long distance communications**
 - a. Medium
 - b. High
 - c. Low
9. **Electron density**
 - a. In E (kennelly heavyside) layer is greater than F layer
 - b. In E layer is less than F layer
 - c. Is same in both the layers
10. **Is radio signal takes 5 micro second to travel from A to B, distance from A to B is:**
 - a. 10,000 m
 - b. 50,000m
 - c. 1500m
11. **Radio waves travelling from sea to land**
 - a. Bends towards the normal
 - b. Bends away from the normal
 - c. Does not bend at all
12. **Duct Propagation -**
 - a. Decreases the VHF range
 - b. Increases the VHF range
 - c. Does not affect the VHF range
13. **If a radio wave is describes a horizontally polarised, in what planes are the i) magnetic and ii) electrical fields oriented?**

- a. i) horizontal ii) horizontal
- b. i) horizontal ii) vertical
- c. i) vertical ii) horizontal

14. Which emission designation would apply to a single channel amplitude modulated single sideband signal with a suppressed carrier wave carrying analogue information from aural (telephonic) reception?

- a. A2A
- b. A8W
- c. J3E

15. Which of the following facilities are most likely to use frequency modulated signal?

- a. Ground-ground UHF radiotelephony
- b. Air-air VHF radio telephony
- c. Ground-air HF data link

16. What is a hertz?

- a. The wavelength in cycles per second
- b. A frequency of one cycle per second
- c. The wavelength corresponding to one cycle per second

17. If wavelength is 8mm, what be frequency?

- a. 37.5 GHZ
- b. 375 GHZ
- c. 3750 GHZ

18. For a frequency of 200 khz, what is the wavelength?

- a. 1500 m
- b. 150 m
- c. 1500 km

19. An AM signal at frequency f_m carried on a transmitted frequency produces:

- a. One sideband of transmission at $f_c + 2f_m$
- b. Two sidebands of transmission at $f_c + f_m$ and $f_c - f_m$
- c. Two sidebands of transmission at $f_c + 2f_m$ and $f_c - 2f_m$

20. What happens to the direction and speed of propagation of radio waves as they pass through air whose density is increasing?

- a. The waves bend towards the high density and speed up
- b. The waves bend towards the high density and speeds down
- c. The waves bend away from the high density and speed up

- 21. If you are using HF voice communications, and a good two-way signal during the day, what must you do as night falls to maintain communications?**
- a. Nothing
 - b. Change to a higher frequency
 - c. Change to a lower frequency,
- 22. What happens in practical terms to the layers of the ionosphere at night?**
- a. The D layer thickens and E layer rises
 - b. The D layer thins and E layer rises
 - c. The E layer thickens and D layer rises
- 23. If the transmission power of a signal sent by both the surface wave and sky wave is doubled, what will happen to the dead space?**
- a. The dead space will reduce from the transmission end
 - b. The dead space will reduce from both the ends
 - c. The dead space will reduce as skip distance reduces
- 24. If the transmission frequency of a signal sent by both the surface wave and sky wave is doubled, what will happen to the dead space?**
- a. Nothing
 - b. The dead space will increase from both the ends
 - c. The dead space will reduce as skip distance reduces
- 25. A single sideband transmission:**
- a. Normally transmits the carrier wave also
 - b. Is used to reduce the atmospheric noise received
 - c. Has an emission designator of E5J
- 26. At what maximum range would a VHF signal transmitted from an ac flying at 14000 ft be received by another ac flying at 9000 ft?**
- a. 190km
 - b. 212km
 - c. 494km
- 27. When may a crew use SELCAL facility of an HF set?**
- a. When ATC publish the facility is available
 - b. When the crew specifically obtains ATC permission to use it
 - c. Both (a) and (b)
- 28. Which of the following is true about satellite communications for aircraft?**
- a. Speech is digitized several conversations can share one frequency
 - b. Speech is digitized; each conversation has its own frequency
 - c. Each satellite uses one frequency for all communications

29. What wavelength is most likely to be used for voice communications over 1000 miles during day?
- 20 cm
 - 2 m
 - 20 m
30. Which wave is used for VHF communications?
- Sky wave
 - Space wave
 - Surface wave
31. On a VDF letdown, the controller passes a QDM of 127 deg class bravo. The 'class bravo' means that the bearing is accurate to within -
- ± 1 deg
 - ± 2 deg
 - ± 5 deg
32. If a pilot wishes Air traffic control to give him headings to steer during a let-down using VHF direction finding. What does he asks for?
- QNH
 - QTE
 - QGH
33. If an aircraft is at 8000 ft, and ground station is at sea level, what will be the maximum range at which the pilot can expect a QTE from the ground station?
- 80nm
 - 107nm
 - 144nm

Answers

1	2	3	4	5	6	7	8	9	10	11
C	B	B	B	A	B	C	B	B	C	A

12	13	14	15	16	17	18	19	20	21	22
B	C	C	C	A	A	A	B	B	C	B

23	24	25	26	27	28	29	30	31	32	33
A	B	B	C	C	A	C	B	C	C	B

Modulation QUESTIONS

1. The bandwidth produced when a radio frequency (RF) of 4716 kHz is amplitude modulated with an audio frequency (AF) of 6 kHz is:
 - a. 6 kHz
 - b. 3 kHz
 - c. 12 kHz
 - d. 9 kHz

2. Which of the following statements concerning AM is correct?
 - a. the amplitude of the RF is modified by the frequency of the AF
 - b. the amplitude of the RF is modified by the amplitude of the AF
 - c. the frequency of the RF is modified by the frequency of the AF
 - d. the frequency of the RF is modified by the amplitude of the AF

3. Which of the following is an advantage of single sideband (SSB) emissions?
 - a. More frequencies available
 - b. Reduced power requirement
 - c. Better signal/noise ratio
 - d. All of the above

4. Which of the following statements concerning FM is correct?
 - a. the amplitude of the RF is modified by the frequency of the AF
 - b. the amplitude of the RF is modified by the amplitude of the AF
 - c. the frequency of the RF is modified by the frequency of the AF
 - d. the frequency of the RF is modified by the amplitude of the AF

ANSWERS

1	2	3	4
C	B	C	D

VDF QUESTIONS

1. An aircraft has to communicate with a VHF station at a range of 300 nm, if the ground station is situated 2,500' amsl which of the following is the lowest altitude at which contact is likely to be made?
 - a. 190'
 - b. 1,378'
 - c. 36,100'
 - d. 84,100'
2. Class 'B' VHF DF bearings are accurate to within:
 - a. $\pm 1^\circ$
 - b. $\pm 5^\circ$
 - c. $\pm 2^\circ$
 - d. $\pm 10^\circ$
3. A VDF QDM given without an accuracy classification may be assumed to be accurate to within:
 - a. 2 degrees
 - b. 5 degrees
 - c. 7.5 degrees
 - d. 10 degrees
4. An aircraft at altitude 9,000 feet wishes to communicate with a VHF/DF station that is situated at 400 feet amsl. What is the maximum range at which contact is likely to be made?
 - a. 115nm
 - b. 400nm
 - c. 143nm
 - d. 63.5nm
5. An aircraft is passed a true bearing from a VDF station of 353° . If variation is 8°E and the bearing is classified as 'B' then the:
 - a. QDM is $345^\circ \pm 5^\circ$
 - b. QDR is $345^\circ \pm 2^\circ$
 - c. QTE is $353^\circ \pm 5^\circ$
 - d. QUJ is $353^\circ \pm 2^\circ$
6. An aircraft at 19,000ft wishes to communicate with a VDF station at 1,400ft amsl. What is the maximum range at which contact is likely?
 - a. 175nm
 - b. 400.0nm
 - c. 62.5nm
 - d. 219nm

ANSWERS

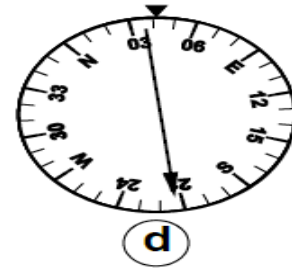
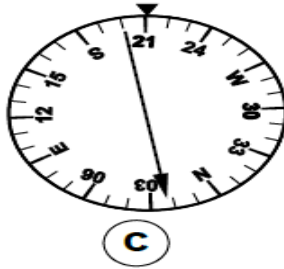
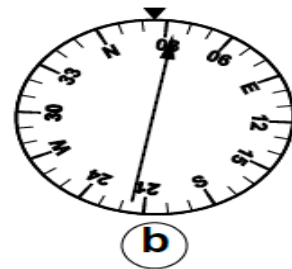
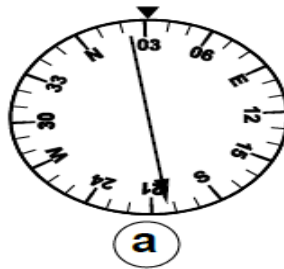
1	2	3	4	5	6
C	B	B	C	C	D

NDB/ADF QUESTIONS

1. **The phenomenon of coastal refraction which affects the accuracy of ADF bearings:**
 - a. is most marked at night.
 - b. can be minimised by using beacons situated well inland.
 - c. can be minimised by taking bearings where the signal crosses the coastline at right angles.
 - d. is most marked one hour before to one hour after sunrise and sunset.
2. **An aircraft is intending to track from NDB 'A' to NDB 'B' on a track of 050° (T), heading 060° (T). If the RBI shows the relative bearing of 'A' to be 180° and the relative bearing of 'B' to be 330° then the aircraft is:**
 - a. Port of track and nearer 'A'.
 - b. Port of track and nearer 'B'.
 - c. Starboard of track and nearer 'A'.
 - d. Starboard of track and nearer 'B'.
3. **ADF Quadrantal Error is caused by:**
 - a. static build up on the airframe and St. Elmo's Fire.
 - b. the aircraft's major electrical axis, the fuselage, reflecting and re-radiating the incoming NDB transmissions.
 - c. station interference and/or night effect.
 - d. NDB signals speeding up and bending as they cross from a land to water propagation path.
4. **The overall accuracy of ADF bearings by day within the Promulgated Range (DOC) is:**
 - a. $\pm 3^\circ$
 - b. $\pm 5^\circ$
 - c. $\pm 6^\circ$
 - d. $\pm 10^\circ$
5. **In order to Tune, Identify and Monitor NON A1A NDB emissions the BFO should be used as follows:**

Tune	Identify	Monitor
a. On	On	Off
b. On	On	On
c. On	Off	Off
d. Off	Off	Off
6. **The magnitude of the error in position lines derived from ADF bearings that are affected by coastal refraction may be reduced by:**
 - a. selecting beacons situated well inland.
 - b. only using beacons within the designated operational coverage.
 - c. choosing NON A2A beacons.
 - d. choosing beacons on or near the coast.

7. An aircraft is tracking away from an NDB on a track of $023^\circ(\text{T})$. If the drift is 8° port and variation 10° west, which of the RMIs illustrated below shows the correct indications?



8. The BFO facility on ADF equipment should be used as follows when an NDB having NON A1A type emission is to be used:
- BFO on for tuning and identification but may be turned off for monitoring.
 - BFO on for tuning but can be turned off for monitoring and identification purpose.
 - BFO off during tuning, identification and monitoring because this type of emission is not modulated.
 - BFO should be switched on for tuning, ident and monitoring.
9. The Protection Ratio of 3:1 that is provided within the Promulgated range/Designated Operational Coverage of an NDB by day cannot be guaranteed at night because of:
- Long range skywave interference from other transmitters.
 - Skywave signals from the NDB to which you are tuned.
 - The increased skip distance that occurs at night.
 - The possibility of sporadic E returns occurring at night.
10. Each NDB has a range promulgated in the COMM section of the Air Pilot. Within this range interference from other NDBs should not cause bearing errors in excess of:
- day ± 5
 - night ± 10
 - day ± 6
 - night ± 5
11. The range promulgated for all NDBs is:
- Within which a protection ratio of 3:1 is guaranteed by day and night.
 - Up to which bearings can be obtained on 95% of occasions.
 - Within which bearings obtained by day should be accurate to within 5° .

- d. Within which protection from skywave protection is guaranteed.
- 12. In order to resolve the 180° directional ambiguity of a directional LOOP aerial its polar diagram is combined with that of a SENSE aerial to produce a..... whose single null ensures the ADF needle moves the shortest distance to indicate the correct.....**
- at the aircraft, cardioid, radial.
 - at the transmitter, limacon, bearing.
 - at the aircraft, limacon, bearing.
 - at the aircraft, cardioid, bearing.
- 13. The protection ratio of NDBs within the Promulgated range (DOC) applies:**
- by day only.
 - by night only.
 - both day and night.
 - at dawn and dusk.
- 14. The phenomena of coastal refraction affecting ADF bearings is caused by the signal _____ when it reaches the coastline and bending _____ the normal to the coast:**
- accelerating towards
 - decelerating towards
 - accelerating away from
 - decelerating away from
- 15. In an ADF system, night effect is most pronounced:**
- during long winter nights.
 - when the aircraft is at low altitude.
 - when the aircraft is at high altitude.
 - at dusk and dawn.
- 16. When the induced signals from the loop and the sense antenna are combined in an ADF receiver, the resultant polar diagram is:**
- a limacon
 - a cardioid
 - figure of eight shaped
 - circular
- 17. When flying over the sea and using an inland NDB to fix position with a series of position lines, the plotted position in relation to the aircraft's actual position will be:**
- further from the coast.
 - closer to the coast.
 - co-incident.
 - inaccurate due to the transmitted wave front decelerating.

18. An aircraft on a heading of $235^{\circ}(\text{M})$ shows an RMI reading of 090° with respect to an NDB. Any quadrantal error which is affecting the accuracy of this bearing is likely to be:
- a maximum value.
 - a very small value.
 - zero, since quadrantal error affects only the RBI.
 - zero, since quadrantal error affects only the VOR.
19. The principal propagation path employed in an NDB/ADF system is:
- skywave
 - surface wave
 - direct wave
 - ducted wave
20. The ADF of an aircraft on a heading of $189^{\circ}(\text{T})$ will experience the greatest effect due to Quadrantal Error if the NDB bears:
- $234^{\circ}(\text{T})$
 - $279^{\circ}(\text{T})$
 - $225^{\circ}(\text{T})$
 - $145^{\circ}(\text{T})$

Answers

1	2	3	4	5	6	7	8	9	10
C	D	B	B	B	D	D	D	A	A

11	12	13	14	15	16	17	18	19	20
C	D	A	C	D	B	B	A	B	A

NDB Numerical

1. Aircraft en-route place A to place B reads on ADF relative bearing 188 on tracking out from A and 008 while tracking in to place B.
 - a. a/c is on track
 - b. a/c is port of track
 - c. a/c starboard of track
2. Total distance 135 nm, ADF reads 170' at 90 nm. Find correction applied to destination.
 - a. 30 port
 - b. 30 starboard
 - c. 15 port
3. An a/c on hdg [T] 015 to maintain track 010° finds ADF, RBI reading 169° at 120 NMS outbound. Calculate-
 - a) distance off track
 - b) hdg correction to destination 60 NMS away.
4. An a/c on hdg [T] 015° to maintain track 010° finds ADF RBI reading 187° at DME 120. Calculate distance off track and heading correction to destination 120 NMS away.
5. An a/c on outbound track from delhi is on a heading to compensate 10°S drift its ADF tuned to delhi NDB should read?
 - a. 190°
 - b. 010°
 - c. 180°
6. A/C on a magnetic bearing of 060° from place P measure brg [R] 310° of NDB at P, if dev is 2°W and Var 3°E. Calculate a) hdg [T] b) hdg[C] c) QTE d) QDM
7. An ac has crossed over place P and is now maintaining hdg [M] 060° with variation 2°W and 8°S drift. Calculate a) TMG [T] b) QTE c) QOM d) ADF reading on RBI.
8. An a/c is on track 300 and heading 300 from A to B total distance 135 nms. After 90 NMs from A, the aircraft ADF tuned to NDB of place A reads 190. The a/c distance off track and heading correction to place are:
 - a. 15 nms and 30S
 - b. 30 nms and 30S
 - c. 15 nms and 30 P
9. An a/c maintaining track, is on radial 105 with heading 120 (T) with variation 8° E. Find relative bearing indicated on ADF.
 - a. 183°

- b. 173°
 - c. 187°
10. Wingtip bearing change= 10° elapsed time between bearings: 8 min. Rate of fuel consumption= 11.8 GPH. The fuel required to fly to the station is?
- a. 8.2 gal
 - b. 9.4 gal
 - c. 15.7 gal
11. Determine approx time and distance to NDB stn if 5° bearing change occurs in 1.5 min with a TAS of 95 kts.
- a. 11 min and 40 nm
 - b. 16 min and 14.3 nm
 - c. 18 min and 28.5 nm
12. An a/c is tracking out on ADF on track 120[T] and is asked to track out on radial 106 under nil wind conditions var 4° E calculate for 30° intercept angle
- a. Initial hdg [T]
 - b. ADF reading at the time of intercepting new track
13. An a/c is tracking in to a NDB maintaining hdg 040(M) and is asked to track in on radial 210. If intercept angle is 30 calculate a) initial hdg [M] b) ADF reading on intercepting new track.
14. The rel. Bearing of station changes from 270 to 260 in 2.5 min the time to station is _____ min.
15. An ac on hdg [M] 170° ADF reading on fixed card 345° from NDB to station A is asked to intercept and maintain radial 355° . If intercept angle= 45° calculate initial hdg [M] and ADF reading on –
- a. 090/140
 - b. 175/310
 - c. 130/045
16. An a/c is flying a track inbound parallel to airway direction of 090 [M], compensating 5 'S drift and is 12 nms starboard of track. What will ADF read when tuned to destination NDB 45 nms away?
- a. 011°
 - b. 349°
 - c. 191°
17. An a/c on track [M], 360, drift 10° by winds from East is crossing a NDB abeam track on its right, The ac hdg [M] and ADF bearing [R] is
- a. 010/100

- b. 010/80
- c. 350/100

18. The a/c tracking in on stn 'A' on radial 200 is asked to intercept and proceed outbound on radial 330. If intercept angle is 40° calculate initial hdg to turn.

- a. 020
- b. 040
- c. 010

Answers

1	A
2	B
3	12nm, 18.5°
4	24, 24° P
5	190°
6	293, 292, 063, 240
7	66° , 66° , 248, 188°
8	C
9	B
10	B
11	C
12	080, 210
13	060° , 330
14	15 min
15	C
16	B
17	B
18	C

VOR Questions –

1. The Designated Operational Coverage quoted for VOR beacons in the COMM section of the Air Pilot:

- a. Is only applicable by day.
 - b. Guarantees a Protection Ratio of at least 3 to 1 by day and night.
 - c. Defines the airspace within which an aircraft is assured of protection from interference from other VORs on the same channel.
 - d. Is determined by the type of surface over which the signal will have to travel.
- 2. What is the theoretical maximum range that an aircraft at flight level 360 will obtain from a VOR beacon situated at 900 feet above mean sea level?**
- a. 274 nms
 - b. 255 nms
 - c. 112 nms
 - d. 224 nms
- 3. A Conventional VOR:**
- a. has an FM reference signal and an AM variable signal
 - b. has a 150Hz reference signal and a 90Hz variable signal
 - c. has an AM reference signal and a 150 Hz variable signal
 - d. has an AM reference signal and an FM variable signal.
- 4. The OBS on a deviation indicator is set to 330° and gives a 3 dots fly right demand with FROM indicated. What is the QDM of the aircraft to the station?**
- a. 144
 - b. 324
 - c. 336
 - d. 156
- 5. An aircraft is homing towards a VOR which marks the centre line of an airway. The beacon is 100 nms distant. If the pilot had the airway QDM set on the OBS what deflection of the deviation indicator would be given if the aircraft was on the boundary of the airway? Assume one dot equals 2 degrees.**
- a. 3 dots
 - b. 2 dots
 - c. 2.5 dots
 - d. 1.5 dots
- 6. What is the theoretical maximum range that an aircraft at flight level 420 will obtain from a VOR beacon situated at 400 feet above mean sea level?**
- a. 225 nm
 - b. 256 nm
 - c. 281 nm
 - d. 257 nm
- 7. Concerning conventional and Doppler VORs (DVOR), which of the following is correct?**
- a. There is no way of knowing from the instrumentation display which type is being used.

- b. The DVOR will always have a "D" in the ident.
 - c. The DVOR has a higher pitch ident than the standard VOR.
 - d. The conventional VOR has less site error.
8. In a Doppler VOR (DVOR) the reference signal is _____, the bearing signal is _____ and the direction of rotation of the bearing signal is _____:
- a. AM, FM, anti-clockwise.
 - b. AM, FM, clockwise.
 - c. FM, AM, anti-clockwise.
 - d. FM, AM, clockwise.
9. An aircraft is attempting to home to a VOR on the 064 radial. The CDI shows 4 dots fly right with a TO indication. At the same time the co-located DME shows a range of 45 nm. Where is the aircraft in relation to the required track?
- a. 6 nm right of track
 - b. 3 nm right of track.
 - c. 6 nm left of track.
 - d. 3 nm left of track.
10. A VOR beacon ceases to transmit its normal identification which is substituted by 'TST'. This means that:
- a. The beacon may be used providing that extreme caution is used.
 - b. The beacon is undergoing maintenance or calibration and should not be used.
 - c. This is a temporary short range transmission and will have approximately half its normal range.
 - d. The beacon is under test and pilots using it should report its accuracy to air traffic control.
11. What is the approximate maximum range that an aircraft flying at 25000' would expect to obtain from a VOR beacon situated 900' above mean sea level?
- a. 220nm
 - b. 100nm
 - c. 235nm
 - d. 198nm
12. An aircraft is on the airway boundary range 100 nm from a VOR marking the airway centre line. Assuming that each dot equates to 2° how many dots deviation will be shown on the deviation indicator?
- a. 3.0 dots
 - b. 2.5 dots
 - c. 2.0 dots
 - d. 1.5 dots
13. An aircraft is required to intercept and home to a VOR along the 064 Radial. The OBS should be set to:

- a. 064 to get correct needle sense and a TO indication.
 - b. 244 to get correct needle sense and a TO indication.
 - c. 064 to get correct needle sense and a FROM indication.
 - d. 244 to get correct needle sense and a FROM indication.
- 14. The type of emission radiated by a VOR beacon is:**
- a. A double channel VHF carrier with one channel being amplitude modulated and the second channel being frequency modulated.
 - b. A single channel VHF carrier wave amplitude modulated at 30 Hz with a sub carrier being frequency modulated at 30 Hz.
 - c. A VHF carrier wave with a 90 Hz frequency modulation and a 150 Hz amplitude modulation.
 - d. A VHF pulse modulated emission with a pulse repetition frequency of 30 pps.
- 15. An aircraft wishes to track towards a VOR along the 274 radial. If variation is 10° W what should be set on the OBS?**
- a. 274
 - b. 264
 - c. 094
 - d. 084
- 16. An aircraft is tracking away from a VOR on a heading of 287°M with 14° starboard drift. If the variation is 6°W what is the phase difference between the reference and variable phase components of the VOR transmission?**
- a. 121°
 - b. 295°
 - c. 301°
 - d. 315°
- 17. What is the theoretical maximum range that a pilot would obtain from a VOR situated 900 feet above mean sea level in an aircraft flying at 18,000 feet?**
- a. 168nm
 - b. 188nm
 - c. 205nm
 - d. 250nm
- 18. An aircraft is attempting to home to a VOR beacon. The pilot has set 329 on the OBS of the deviation indicator. If the aircraft is situated on the 152 radial then the deviation indicator will show:**
- a. one and a half dots fly right.
 - b. one and a half dots fly left.
 - c. three dots fly right.
 - d. three dots fly left.
- 19. A VOR receiver in an aircraft measures the phase difference from a DVOR as 220°. Which radial is the aircraft on?**
- a. 140
 - b. 040
 - c. 320
 - d. 220
- 20. The RMI indicates the aircraft magnetic heading. To convert the RMI bearings of NDBs and VORs to true bearings, the correct combination for the application of magnetic variation is:**
- | | |
|-----|-----|
| NDB | VOR |
|-----|-----|

- | | |
|----------------------|-------------------|
| a. beacon position | aircraft position |
| b. beacon position | beacon position |
| c. aircraft position | beacon position |
| d. aircraft position | aircraft position |

21. The normal maximum error which might be expected with a VOR bearing obtained within the DOC is:

- a. plus or minus 1°
- b. plus or minus 2°
- c. plus or minus 5°
- d. plus or minus 10°

22. The VOR indications on an RMI whose deviation is not zero:

- a. are magnetic.
- b. are compass.
- c. are relative.
- d. must have deviation applied before being used.

Answers

1	2	3	4	5	6	7	8	9	10	11
C	A	A	A	D	C	A	A	C	B	C

12	13	14	15	16	17	18	19	20	21	22
D	B	B	C	C	C	A	D	C	C	A

VOR Numerical

1. An a/c on heading 090[M] with OBS selected to 060. The ac is on bearing 040[M] from the stn. The TO/FROM indication and CDI needle will show?
 - a. From and fly right
 - b. From and fly left.
 - c. To and fly right

2. If a/c hdg is 070° [M] and DME reads 90, OBS, 062° and indication on 4 dot indicator is 2 dots fly right and flag 'TO'. Find -
 - a. Aircraft present radial
 - b. Distance off track
 - c. ADF fixed card indication from NDB co-located

3. An a/c is on hdg 110° [M] and on brg [M] 295° from the VOR station. The DME reads 60 nms VOR indications pm at OBS 290°. Find.
 - a. RMI readings

- b. CDI indications on a 5 dot indicator.
4. An a/c is on inbound track on radial 040° selects OBS 055° and turns 15° port. From this point CDI needle takes 15 Min to centre. Calculate his time away from the VOR station.
 5. An a/c on radial 0904. An a/c is on outbound track on radial 040° selects OBS 055° and turns 15° port. From this point CDI needle takes 15 Min to centre. Calculate his time away from the VOR station.
 6. An a/c on hdg 050°[T] is tracking in on radial with the OBS setting 068 and TO/FROM indicates 'to' variation is 10°W. Find ADF bearing from a NDB co-located with VOR.
 7. At DME 60 on inbound track VOR 4 dot indicator shows 2 dot fly left. OBS selected is 270° and hdg [M] is 270° find
 - a. Distance off track
 - b. QDR
 - c. TO/FROM indication TO
 - d. Indication of ADF co-located
 8. An a/c is tracking in on radial 274° to a VOR station. Drift 10°S, Variation at grnd stn 5° W and a/c DR position 3°W. Calculate heading to fly.
 - a. 104° [M]
 - b. 084° [M]
 - c. 094° [T]
 - d. 087° [M]

Answers

1	2	3	4	5	6	7	8
A	247,7.5NM, 357	110 & 115, 2.5 from left	15 min	120nm	008°	5nm, 085,TO, 355	B

RMI QUESTIONS

1. An a/c is on radial 158 with OBS 162
 - a. What is VOR indication on 5 dot indicator.
 - b. ADF indication on RMI/ RBI if a/c hdg [M] is 162

- c. ADF indication on RMI/RBI if hdg [M] is 342
2. An a/c on hdg 041 [M] and 20 nms from VOR is on brg of 215 [M] from VOR stn. If OBS selected is 212 and TO/ FOM indication is 'FROM' find indication on -
 - a. RMI
 - b. CDI OF ILS/VOR S dot indicator
 - c. RBI reading of co-located NDB
 3. An a/c on radial 250 from place P with OBS 250 and 'from' indication with 2°E variation observes bearing [R] 200 from NDB at P. Calculate
 - a. Hdg[T]
 - b. RMI reading
 4. An a/c tracking out on Mumbai NDB on hdg 175[M] reads 355° on RMI bearing needle. If var 8° E and dev = 3°E. The TMG [T] of a/c is?
 - a. 183°
 - b. 167°
 - c. 170°
 - d. 355°
 5. An a/c is on hdg 055°, RMI reading 060°. After sometime hdg remains same but RMI reads 057°. The a/c is drifting to -
 - a. Port
 - b. Starboard
 6. An a/c RMI is tuned to VOR and NDB while tracking in on radial 217° to a VOR stn on a hdg compensating for 7°S drift. The NDB bears 055° [rel] from a/c. Calculate RMI readings.
 7. An a/c RMI is tuned to NDB and VOR. The a/c is tracking out on radial 217 from VOR stn compensating for 7°S drift. ADF reads 055° [R]. What are the indication on RMI.
 8. An aircraft on hdg (C) 040 is homing on to station P with Variation 4W and deviation 2W and drift 6P. Calculate –
 - a. ADF reading on RBI
 - b. RMI reading
 - c. QTE from P

Answers

1	2	3	4	5	6	7	8
2 dots R, 338°, 176°, 356°	035, 1.5L, 354	232 (T); 070	A	B	037; 030; 085	210°, 265°, 037°	354°, 32°, 208°

VOR Entries

1. The DPN NDB hold is a right hand pattern, the inbound track being 031°. You are required to track towards DPN on a QDM 230° and enter the hold. The sector entry will be:

- 2. The PLG hold is a standard hold, the inbound track being 084°. Your present heading to the beacon is 010°. The sector entry will be:**

 - a. Direct
 - b. Parallel
 - c. Offset
- 3. The BBB hold is a non-standard pattern, the inbound track being 258°. You are required to track inbound on the 270° dial and hold. The sector entry will be?**

 - a. Direct
 - b. Parallel
 - c. Offset
- 4. The MMV hold is a left hand pattern the outbound track of the pattern being 247°. ATC request that you track inbound on the 050° radial and enter the hold. The sector entry will be?**

 - a. Direct
 - b. Parallel
 - c. Offset
- 5. The holding pattern at DPN is non-standard. The inbound track of the pattern is 236°. Your present heading to DPN is 126° whilst making good a track of 125°. To enter the hold your sector entry will be:**

 - a. Direct
 - b. Parallel
 - c. Offset
- 6. You are required to enter the hold at CTV. It is a left hand hold with an inbound track of 008°. Your track towards the VOR is 076° with 2° left drift. Your sector entry will be:**

 - a. Direct
 - b. Parallel
 - c. Offset

1	2	3	4	5	6
B	B	C	B	B/C	A/C

1. The coverage of an ILS localiser extends to _____ either side of the on course line out to a range of _____ nm.

a. 10°, 35

b. 35°, 10

- c. 35° , 17
 - d. 25° , 25
2. **The upper and lower limits of an ILS glide path transmitter having a 3.5° glide slope are:**
 - a. 6.125° - 1.575°
 - b. 7.700° - 1.225°
 - c. 5.250° - 1.350°
 - d. 3.850° - 3.150°
 3. **The minimum angle at which a false glide path is likely to be encountered on a 3° glide path is:**
 - a. 6 degrees
 - b. 5.35 degrees
 - c. normal glide slope times 1.75
 - d. normal glide slope times 0.70
 4. **The visual and aural indications obtained when overflying an ILS middle marker are:**
 - a. continuous low pitched dashes with synchronised blue light.
 - b. continuous high pitched dots with synchronised amber light.
 - c. alternating medium pitch dots and dashes with amber light.
 - d. one letter in Morse with synchronised white light.
 5. **An aircraft carrying out an ILS approach is receiving stronger 150 Hz signals than 90 Hz signals. The correct actions to be taken to place the aircraft on the centreline and on the glide path are to fly:**
 - a. DOWN and LEFT.
 - b. UP and LEFT
 - c. UP and RIGHT.
 - d. DOWN and RIGHT.
 6. **In elevation the upper and lower limits of an ILS glide path transmitter having a 3.0° glide slope are:**
 - a. 0.35° 0.70°
 - b. 3.000° at least 6°
 - c. 5.25° 1.35°
 - d. 10.0° 35.0°
 7. **A category 2 ILS installation:**
 - a. provides accurate guidance down to 50' above the horizontal plane containing the runway threshold.
 - b. has a steep glide path, normally 7.5° .

- c. provides accurate guidance down to the runway and along the runway after landing.
 - d. has a false glide path that is exactly twice the true glide path angle.
- 8. The outer marker of an ILS installation has a visual identification of:**
- a. alternating dots and dashes on a blue light.
 - b. continuous dots at a rate of 3 per second, blue light.
 - c. continuous dashes at a rate of 2 per second, amber light.
 - d. continuous dashes at a rate of 2 per second, blue light.
- 9. The specified maximum safe fly up indication on a 5 dot CDI is:**
- a. half full scale needle deflection above the centre line.
 - b. 2.5 dots fly up.
 - c. just before full scale deflection.
 - d. 1.3 dots fly up.
- 10. The coverage of the ILS glideslope in azimuth is:**
- a. $\pm 8^\circ$ out to 10nm
 - b. $\pm 10^\circ$ out to 8nm
 - c. $\pm 12^\circ$ out to 17nm
 - d. $\pm 35^\circ$ out to 25nm
- 11. An a/c is 6nm from touchdown on ILS approach. The GP needle (4 dot) is one dot below the centre line. While on GS of 3° . What is the height of the a/c?**
- a. 1924 ft
 - b. 1654ft
 - c. 2056 ft
 - d. 1880ft
- 12. Glideslope on an ILS approach operates on –**
- a. 108 MHz
 - b. 332 MHz
 - c. 512 MHz
- 13. Where would you expect to find a false glideslope on an ILS?**
- a. On the localiser, below the correct glide slope
 - b. On the back beam, below the correct glide slope
 - c. On the localiser, above the correct glide slope
 - d. On the back beam, above the correct glide slope
- 14. Which frequency bands are transmitted by ILS equipment?**
- a. Glideslope on VHF, Localiser on VHF
 - b. Glideslope on VHF, Localiser on UHF
 - c. Glideslope on UHF, Localiser on VHF
 - d. Glideslope on UHF, Localiser on UHF
- 15. An ILS glideslope signal may be received with the following footprint:**
- a. 8° either side of the centerline, out to 25nm

- b. 8° either side of the centerline, out to 10nm
- c. 10° either side of the centerline, out to 25nm
- d. 10° either side of the centerline, out to 10nm

16. Category III ILS allows approaches with following criteria –

- a. DH below 30m, appropriate RVR.
- b. DH below 60m, RVR below 350m.
- c. DH below 30m, RVR 350m.
- d. DH below 60m, RVIZ 550m.

17. The frequency band 108 – 112 MHz contains –

- a. VOR signals
- b. ILS signals
- c. VOR and ILS signals
- d. VOR and VHF communication signals.

18. What form of emission does an ILS signal use?

- a. A2A
- b. A3E
- c. A8W
- d. J3E

Answers

1	2	3	4	5	6	7	8	9
C	A	A	C	B	C	A	D	B

10	11	12	13	14	15	16	17	18
A	A	B	C	C	B	A	C	C

ILS NUMERICAL

1. An a/c is 6 nm from touch down on ILS approach. The GP needle (4 dot) is one dot below the centre line. While on glideslope of 3 deg. What is the height of the a/c?
2. An a/c on ILS has a half full scale deflection on the GP pointer. Giving fly up indication of 2.5 nm. Range from touchdown, there is obstruction of 262 ft above threshold. What is the vertical clearance of a/c if the G slope of ILS is 2.8 deg?
3. At a distance of 6 nm from the touchdown pilot receives full deflection of GP indicating fly up. Approximately how many feet below the GP the ac is at this time?
4. While approaching the airfield you are above the glidepath and to the left of the centreline of the R/W. The indication on the ac will be
 - a. Below and to the left
 - b. Above and to the left
 - c. Below and to the right
5. An a/c is settled on the ILS both LOC and GP needles showing in the centre. The ILS glide slope is 2.5 deg. What is the height of the ac when it is over MM 1.5nm from the touchdown?
6. When an ILS indication the glide path needle is below and the localiser to the left. The a/c is:
 - a. Above the glide path to the right of the localiser
 - b. On the glide path to the right of the localiser
 - c. On the glide path and localiser abeam
7. Glideslope on an ILS approach operates on?
 - a. 108MHZ
 - b. 332MHZ
 - c. 500MHZ
8. Difference between approach azimuth and back azimuth is:-
 - a. MLS can be used in reverse sensing in back azimuth
 - b. Back azimuth does not have the 'DME P'
 - c. The back azimuth transmits at the lower level/rate
9. To remain on an ILS glideslope the rate of descend must be
 - a. Increased if the airspeed is increased
 - b. Decreased if the airspeed is increased
 - c. Decreased if the ground speed is increased

- d. Increased if the ground speed is increased
- 10. On an ILS approach if an a/c is too high on glideslope and still opts to land. It may cause?**
- Hard landing
 - Lower landing roll
 - Touch down short of runway
- 11. The outer marker of an ILS system emits-**
- Blue flashes synchronized with audio dashes
 - White flashes synchronized with audio dashes
 - Blue flashes synchronized with audio dots
 - White flashes synchronized with audio dots
- 12. Up to what angular distance from the centreline would correct MLS azimuth indications?**
- 20° either side
 - 30° either side
 - 40° either side
- 13. When an a/c is on localizer indications needle to the full left on the final approach, how many degrees deflection and which side to turn to come back to the localizer?**
- 2.5° and left
 - 2.5° and right
 - 0.7° and left
- 14. After takeoff from ILS the localizer needle is in the yellow sector, the correction to reach the centreline of r/w will be toward?**
- Right
 - Left
 - Cannot be determined because the localiser works only on the final approach
- 15. Height of the glideslope line at the middle marker is _____ .**
- 16. What is the lateral approach azimuth angles limits to either side of the landing runway of MLS.**
- 17. On a flight director system, if a GS flag shows it indicates**
- Ground speed being fed to the instrument is not correct
 - Glideslope Rx failure or malfunction
 - Glideslope Tx on ground has failed or malfunctioning
 - Both b and c are correct

18. An a/c on ILS GP 3.25° at 4.5nm GS 140kt, height of ac at outer marker is ____ .
19. On a 3° GS angle a head wind changes to tailwind what will be effect on the ac?
20. While going over the outer marker the indications are ____ .
21. Where would you expect to find a false glide slope on an ILS?
- On the localiser, below the correct glide slope
 - On the back beam, below the correct glide slope
 - On the localiser, above the correct glideslope
 - On the back beam, below the correct glide slope
22. Which frequency bands are transmitted by equipment?
- Glideslope on VHF, localizer on VHF
 - Glideslope on VHF, localizer on UHF
 - Glideslope on UHF, localizer on VHF
23. A category 1 ILS allows approaches with the following criteria:
- Decision height below 30m, an appropriate RVR
 - Decision height not below 30m, RVR 350m.
 - Decision height not below 60m, RVR 550 m
24. What ROD should an a/c use on a 3.5° ILS glide slope if has a GS of 120 knots?
- 500 fpm
 - 600 fpm
 - 700 fpm
25. The frequency band 108 Mhz to 112 Mhz contains
- VOR signals only
 - ILS signals only
 - VOR and ILS signals
26. An ILS glideslope signal may be received with the following footprint:
- 8° either side of the centreline, out to 25 nm
 - 8° either side of the centreline, out to 10 nm
 - 10° either side of the centreline, out to 25 nm
27. An ILS glidestope indicator measures:
- The difference in strength between 2 signals of 90 MHZ and 120 MHZ

- b. The difference in phase between 2 signals in the UHF band
- c. The difference in modulation depth of 2 signals

28. What height should an a/c on a 2.5° glideslope be, at a range of 6 nm from touch down, given an ILS reference of 50 feet.

- a. 1350 ft
- b. 1550 ft
- c. 1750 ft

29. What form of emission does an ILS signals

- a. AZA
- b. A3E
- c. A8W

30. A category III ILS allows approaches with the following criteria

- a. Decision height below 30m, an appropriate RVR
- b. Decision height not below 30m, RVR 350m
- c. Decision height not below 60m, RVR 350m

31. What rate of descent should an a/c use on a 2.5° ILS glideslope if it has a GS of 120 knots.

- a. 500 fpm
- b. 600 fpm
- c. 700 fpm

Answers

1	2	3	4	5	6	7	8	9	10	11
1930'	358'	426'	C	380'	A	B	C	D	A	A

12	13	14	15	16	17	18	19	20
C	A	A	175'	40°	C	1482'	Will go above GP	Blue flashes, audio 2 dashes/sec

21	22	23	24	25	26	27	28	29	30	31
C	C	C	C	C	B	C	B	C	A	A

QUESTIONS ILS Numerical

1. An aircraft on ILS approach of standard GP the indication is 3 dots fly down on a 4 dot indicator. If Aircraft is 3.6 DME from threshold at 210 knots ground speed, find -
 - a. Height of aircraft
 - b. ROD for standard GP
 - c. ROD for current position
2. An a/c on 3.25° glide slope is 4.5 NMs from threshold the a/c should be
 - a. 1582 ft
 - b. 1710 ft
 - c. 1482 ft
3. At 3° glide path and at 3 NMS from threshold on ILS approach the radio altimeter should read _____ feet.
4. On a standard GP approach on ILS. If TAS 130 knots, R/W 09, W/V= 090/30. Find ROD.
5. An a/c on ILS approach on standard GP is 3.5 NMs from threshold crossing, a hill 540 ft when radio altimeter reads 400 ft. What will be the GP needle on ILS/VOR indicator read?
6. On standard ILS DME reading 1.9° glideslope needle reads 2 dots below central position on 4 dot indicator. Radio altimeter should read-
 - a. 6450 ft
 - b. 645 ft
 - c. 564 ft

Answers

1	1286; 1050; 1234
2	C
3	912 ft
4	500 ft/min
5	$\frac{1}{2}$ Full deflection fly up
6	645 ft

RADAR QUESTIONS

1. **The factor which determines the maximum range of a radar is:**
 - a. pulse repetition rate
 - b. pulse width
 - c. power
 - d. beamwidth
2. **The main advantage of continuous wave radars is:**
 - a. No maximum range limitation
 - b. Better range resolution
 - c. No minimum range limitation
 - d. Better range resolution
3. **If the PRF of a primary radar is 500 pulses per second, the maximum range will be:**
 - a. 324 nm
 - b. 300 nm
 - c. 162 nm
 - d. 600 nm
4. **To double the range of a primary radar would require the power to be increased by a factor of:**
 - a. 2
 - b. 4
 - c. 8
 - d. 16
5. **The time between the transmission of a pulse and the reception of the echo from a target is 1720 microseconds. What is the range of the target?**
 - a. 139 km
 - b. 258 km
 - c. 278 km
 - d. 516 km
6. **A radar is required to have a maximum range of 100 nm. What is the maximum PRF that will achieve this?**
 - a. 1620 pulses per second (pps)
 - b. 1234 pps
 - c. 617 pps
 - d. 810 pps
7. **If the PRI of a radar is 2100 microseconds, the maximum range of the radar is:**
 - a. 170 nm
 - b. 315 nm
 - c. 340 nm
8. **An echo is received from a target 900 microseconds after the pulse was transmitted. The range to the target is:**
 - a. 73 nm
 - b. 270 nm
 - c. 135 nm
 - d. 146 nm
9. **The factor which limits the minimum detection range of a radar is:**
 - a. pulse repetition interval

- b. transmitter power
 - c. pulse width
 - d. pulse repetition frequency
- 10. The use of Doppler techniques to discriminate between aircraft and fixed objects results in second trace returns being generated. These are removed by:**
- a. using a different frequency for transmission and reception
 - b. jittering the PRF
 - c. making regular changes in pulsewidth
 - d. limiting the power output of the radar
- 11. A radar is designed to have a maximum range of 12 km. The maximum PRF that would permit this is:**
- a. 25000 pps
 - b. 6700 pps
 - c. 12500 pps
 - d. 13400 pps
- 12. The bearing of a primary radar is measured by:**
- a. phase comparison
 - b. searchlight principle
 - c. lobe comparison
 - d. DF techniques
- 13. A ground based radar with a scanner rotation of 60rpm, a beam width in the order of .5° and a PRF of 10000pps would be:**
- a. an Airfield Surface Movement Indicator with a theoretical range of 8nm.
 - b. a Precision Approach Radar.
 - c. an Airfield Surface Movement Indicator with a theoretical range of 16nm.
 - d. a high resolution Surveillance Approach Radar.
- 14. The frequency band and rate of scan of Airfield Surface Movement radars are:**
- a. SHF; 60RPM
 - b. SHF; 200RPM
 - c. EHF; 100RPM
 - d. EHF; 10RPM

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	C	C	D	B	D	A	A	C	B	C	B	A	A

AWR QUESTIONS

1. **A frequency used by airborne weather radar is:**
 - a. 8800 MHz
 - b. 9.375 GHz
 - c. 93.75 Ghz
 - d. 1213 Mhz
2. **An airborne weather radar is required to detect targets up to a maximum range of 200 nm. Ignoring pulse length and flyback in the CRT calculate the maximum PRR.**
 - a. 405 pps
 - b. 810 pps
 - c. 1500 pps
 - d. 750 pps
3. **Using airborne weather radar the weather beam should be used in preference to the fan shaped beam for mapping in excess of _____ nm:**
 - a. 20 to 25
 - b. 60 to 70
 - c. 100 to 150
 - d. 150 to 200
4. **Airborne Weather Radar is an example of radar operating on a frequency of in the band.**
 - a. primary 8800 MHz SHF
 - b. secondary 9.375 MHz UHF
 - c. secondary 9375 MHz SHF
 - d. primary 9375 Mhz SHF
5. **A prominent island is identified on the 30° right bearing line and the 10 nm range marker of an airborne weather radar. If the heading is 045° (T) and the aircraft is at FL360 what range and bearing should be plotted in order to obtain a fix?**
 - a. 10nm 030° (T)
 - b. 10nm 075° (T)
 - c. 8nm 075° (T)
 - d. 8nm 255° (T)
6. **The correct sequence of colours of a colour Airborne Weather Radar as returns get stronger is:**
 - a. red yellow green.
 - b. yellow green red.
 - c. green yellow red.
 - d. red green yellow.
7. **A false indication of water may be given by the AWR display when:**
 - a. flying over land with the Land/Sea switch in the Sea position.

- b. flying over mountainous terrain.
 - c. there is cloud and precipitation between the aircraft and a cloud target.
 - d. attempting to use the mapping beam for mapping in excess of 50 nm.
- 8. An aircraft heading 017° (T) has a small island showing on the AWR at 45nm range on the 60° left azimuth line. To obtain a fix from this information you should plot:**
- a. range 45nm and QTE 060 from the centre of the island.
 - b. range 45nm and QTE 240 from the centre of the island.
 - c. range 45nm and QTE 317 from the centre of the island.
 - d. range 45nm and QTE 137 from the centre of the island.
- 9. An aircraft heading 137° (M) has a small island showing on the AWR at 45nm range on the 30° left azimuth line. Local variation is 12° W. To obtain a fix from this information you should plot:**
- a. Range 45 nm and QTE 095 from the centre of the island.
 - b. Range 45 nm and QTE 275 from the centre of the island.
 - c. Range 45 nm and QTE 107 from the centre of the island.
 - d. Range 45 nm and QTE 287 from the centre of the island.
- 10. Airborne weather radar operates on a frequency of:**
- a. 8800 MHz because gives the best returns from all types of precipitation
 - b. 13300 MHz
 - c. 9375 MHz because it gives the best returns from rainfall associated with Cb
 - d. 9.375 GHz because this frequency is best for detecting aircraft in flight.
- 11. The mapping mode of Airborne Weather Radar utilises:**
- a. a pencil/weather beam from 70 nm to 150 nm.
 - b. a cosecant/fan shaped beam which is effective to 150 nm.
 - c. a pencil/weather beam with a maximum range of 70 nm.
 - d. a cosecant/ fan shaped beam effective 50 nm to 70 nm.
- 12. An Airborne Weather Radar system uses a frequency of 9 GHz because:**
- a. it has a short wavelength so producing higher frequency returns.
 - b. the short wavelength allows signals to be reflected from cloud water droplets of all sizes.
 - c. the wavelength is such that reflections are obtained only from the larger water droplets.
 - d. the frequency penetrates clouds quite easily enabling good mapping of ground features in the mapping mode.
- 13. The antenna of an Airborne Weather Radar is stabilized:**
- a. in pitch, roll and yaw.

- b. in pitch and roll.
 c. in pitch and roll whether the stabilisation is on or off.
 d. in pitch and roll but only when 0° tilt has been selected.
14. The centre of a small island is identified at the intersection of the 60° left bearing line and 15nm range arc of an airborne weather radar. If the aircraft's heading and height are 035° (M) and 42,500ft what QTE and range should be plotted in order to obtain a fix from the island? (variation is 20°W)
- a. 175 15nm
 b. 135 15nm
 c. 135 13nm
 d. 155 14nm
15. The colours used to denote variations in rainfall rate on an Airborne Weather Radar screen are..... for very light or no returns, for light returns,..... for medium returns and for strong returns.
- a. black, yellow, green, magenta.
 b. black, green, yellow, magenta
 c. grey, green, yellow, red.
 d. black, green, yellow, red.
16. The radar in an aircraft at FL370 detects a cloud at 60 nm. The cloud disappears when the tilt is selected to 2° UP. If the beamwidth of the radar is 6°, at what altitude are the tops of the clouds?
- a. 6,000 ft
 b. 31,000 ft
 c. 43,000 ft
 d. 49,000 ft

Answers

1	2	3	4	5	6	7	8
B	A	B	D	D	C	B	D

9	10	11	12	13	14	15	16
B	C	A	B	B	C	D	B

Radio Altimeter Questions

1. The radio altimeter readings will be most reliable at –

- a. 500'
 - b. 1500'
 - c. 5000'
- 2. The radio altimeter suffers from mushing errors –**
- a. at a very great height
 - b. at low heights
 - c. When the Tx and Rx aerials are not at the same level.
- 3. The radio altimeter indicates 2000 ft. What is the minimum height that the aircraft could be on?**
- a. 1935' in the cruising altitude
 - b. 1935' in the landing altitude
 - c. 1898' in the cruising altitude
- 4. A/c flying at FL 250 with its altitude alerting system set to that flight level. If it enters a climb, at what altimeter indication will the altitude alert system warn the pilot?**
- a. 25100'
 - b. 25300'
 - c. 25500'
- 5. Radio Altimeter determines the a/c height by the principle of?**
- a. It measures the doppler shift of a CW transmission.
 - b. It measures the doppler shift of a pulse transmission.
 - c. It measures the time from transmission to reception of CW frequencies.
- 6. What would be the wavelength of a radio altimeter transmission?**
- a. 7mm
 - b. 7cm
 - c. 14cm
- 7. Which of the following information will a radio altimeter give the pilot?**
- a. His DH
 - b. His height above the ground ahead
 - c. His actual height above the runway he is approaching
- 8. How can radio altimeter information be presented to the pilot?**
- a. on a dial with logarithmic scale
 - b. By lights & Aurally
 - c. Both
- 9. Radio altimeters operate in the –**
- a. VHF Band

- b. SHF Band
- c. VLF Band

10. Radio altimeters operate upto –

- a. 2500'
- b. 1500'
- c. 500'

11. The mean frequency on which radio altimeters operate is –

- a. 4200 Mhz
- b. 4300 Mhz
- c. 4400 Mhz

12. A DH setting knob and DH set indicator:

- a. Can only be generated in a radio altimeter if it is a part of EFIS.
- b. Is usually incorporated in modern radio altimeters.
- c. Has to be a separate installation to the radio altimeter itself.

13. The formula used to determine the height displayed on the radio altimeter is that the difference in frequency equals the rate of change of frequency?

- a. Times half the time taken
- b. Times the time taken
- c. Times twice the time taken

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13
A	C	B	B	C	B	A	C	B	A	B	B	B

GPWS QUESTIONS

1. **The Ground Proximity Warning mode 5 provides a visual and audible warning to the pilot if the aircraft:**
 - a. descends below 500ft radio altitude with gear retracted.
 - b. is below 1000ft radio altitude and more than 1.3 dots below the ILS glidepath.
 - c. descend below 200ft radio altitude with flaps retracted.
 - d. sinks more than approximately 10% of accumulated altitude.
2. **The GPWS uses inputs from:**
 - a. the radio altimeter, the ILS receiver, the Air Data Computers and the landing gear position indicators.
 - b. the radio altimeter, the Air Data Computers, the landing gear position indicators and the flap position indicators.
 - c. the radio altimeter, the Air Data Computers, the Captain's ILS receiver, the landing gear position indicators and the flap position indicators.
 - d. the radio altimeter and the ILS receiver.
3. **GPWS Mode One gives warning of:**
 - a. excessive descent rate.
 - b. height loss after take-off/missed approach.
 - c. unsafe terrain clearance when not in the landing configuration.
 - d. excessive terrain closure rate.
4. **GPWS, Mode Three gives warning of:**
 - a. excessive descent rate.
 - b. height loss after take-off/missed approach.
 - c. unsafe terrain clearance when not in the landing configuration.
 - d. excessive terrain closure rate.
5. **In GPWS, BITE can be achieved?**
 - a. In flight only
 - b. On ground only
 - c. At a closed site
 - d. All of the above
6. **GPWS, mode two operates between:**
 - a. 50ft and 2450ft agl
 - b. 50ft and 1800ft agl
 - c. 50ft and 700ft agl
 - d. 50ft and 500ft agl
7. **With reference to GPWS:**
 - a. In all six modes the audible alerts and warnings are accompanied by the red flashing PULL-UP' light.
 - b. Mode 4A activates when the aircraft descends below 500ft radio altitude at a speed less than .35Mach with the landing gear retracted.

- c. Mode 4A activates when the aircraft descends below 500ft barometric altitude at a speed less than .28Mach with the landing gear retracted.
 - d. Mode 6 re-arms when the aircraft leaves the hard alerting area.
- 8. GPWS mode 3 will operate if altitude loss occurred before you have acquired:**
- a. 700 ft barometric altitude gain.
 - b. 500 ft terrain clearance.
 - c. 200 ft barometric altitude gain.
 - d. 700 ft terrain clearance.
- 9. On EGPWS, audio alert indications are given?**
- a. 30 seconds before approaching terrain in red color
 - b. 30 seconds before approaching terrain in yellow color
 - c. 60 seconds before approaching terrain in red color
 - d. 60 seconds before approaching terrain in yellow color
- 10. Mode 4 gives warning of:**
- a. excessive descent rate.
 - b. height loss after take-off/missed approach.
 - c. unsafe terrain clearance when not in the landing configuration.
 - d. excessive terrain closure rate.
- 11. On a GPWS the mode 5 is _____ and it will be inhibited when mode ____ is on?**
- a. Excessive ROD, mode 1
 - b. Excessive ROD, mode 2
 - c. Glide slope, mode 3
 - a. Glide slope, mode 2
- 12. On RA/VSI the eyebrow FLY TO vertical speed indication is shown in which color?**
- a. Green
 - b. Gold
 - & Cross hatched white
- 13. On EGPWS when in audio alert indication given?**
- a. 30 seconds before approaching terrain
 - b. 60 seconds before approaching terrain
 - c. 60 seconds in closing terrain in red colour
 - d. 30 seconds in closing terrain in red colour
- 14. An a/c receives warning on EGPWS. Pilot should -**
- a. Disengage autopilot and initiate recovery action

- b. Contour with the autopilot and initiate recovery action
- c. Look around the maneuvering area before attempting any action
- d. Ask ATC for instructions

15. Mark the correct statements in respect of EGPWS?

- a. It gives indications of approaching wind shear
- b. Navigation indications of approaching wind affecting the a/c
- c. Both a and b are correct
- d. Aural warning of wind shear and a visual warning by illuminating a red light

16. EGPWS gives a caution alert

- a. 60 seconds ahead of terrain conflict
- b. As in A and is repeated every 7 seconds till you are clear of the obstruction.
- c. 20 seconds ahead of terrain conflict
- d. 30 seconds ahead of terrain conflict

17. In EGPWS terrain in magenta colour indicates?

- a. Terrain highly above the a/c altitude
- b. Terrain positively below the a/c altitude
- c. Terrain 500 to 2000 ft above the a/c altitude
- d. No database or unknown terrain

18. EGPWS gives warning to the crew

- a. Only if the terrain display is activated
- b. For terrain besides basic GPWS features
- c. Regardless whether the terrain of t not obstructions are selected or not
- d. Both b and c are correct

19. EGPWS 50% of terrain indications in red indicates:

- a. Unknown terrain
- b. Obstruction is 1000 to 2000 ft above a/c
- c. Obstruction is more than 2000 ft above a/c
- d. Obstruction threat area warning

20. EGPWS mode 5 glide slope warning below 1000 ft radio altitude comes on when-

- a. a/c is 1 dot or more below glide slope
- b. a/c is 1.3 dot or more below glide slope
- c. a/c is 1.5 dot or more below glideslope
- d. a/c is 1.5 dot or more below glideslope

21. What does a GPWS mode 3 alert indicate to a pilot?

- a. The a/c is close to the ground with undercarriage

- b. The a/c is sinking after t/o or go-around
- c. The a/c is below its selected decision height
- d. The a/c is approaching high ground

22. How low on ILS glide slope will the GPWS alert the pilot?

- a. When the a/c is below the bottom of the glide slope
- b. When the a/c is sinking after take-off or go-around
- c. The a/c is below its selected decision height
- d. The a/c is approaching high ground

23. What is the principle of EGPWS?

- a. It combines the radio altimeter with many sensors to compute safe descent rates at varying heights
- b. It sends radar signals ahead of the a/c to sense high ground ahead
- c. It gives an infer-red TV picture of the front path through
- d. It uses GPS and a map database to compute safe clearance from ground ahead

24. Mode 4 on the GPWS give

- a. Warning of terrain clearance between 50 to 500 ft with LG not down
- b. Warning of terrain clearance between 50 to 500 ft with flaps not down
- c. Both are correct

25. In GPWS the BITE can be achieved?

- a. On ground only
- b. In flight only
- c. Both on the ground and in flight

26. Mode 1 on GPWS gives a warning

- a. Below 2450 ft agl
- b. Below 2450 ft amsl
- c. Of excessive rate of descent with respect to terrain
- d. Both A and C are correct

27. Mode 5 of GPWS, mode 5 is inhibited when mode.....is in use

- a. 1
- b. 2
- c. 4

28. At what height GPWS starts warning when the flaps are down?

- a. 790 ft
- b. 2450 ft
- c. 500 ft

- 29. What does a GPWS mode 3 alert indicated to a pilot?**
- a. The a/c is close to the ground with the undercarriage up
 - b. The a/c is sinking after t/o or go around
 - c. The a/c is below its selected decision height
 - d. The a/c is approaching high ground
- 30. Which of the following is not an input to a GPWS**
- a. Radio altimeter
 - b. ILS localiser
 - c. Vertical speed sensor
 - d. Flap position sensor
- 31. Which of the following would not trigger an alert from the GPWS?**
- a. A high sink rate of 1000 agl in the landing configuration
 - b. A high sink rate at 1500ft agl with gear and flaps up
 - c. A low sink rate during go around
 - d. Level flight at 400 ft with undercarriage up
- 32. How long on the ILS glide slope will the GPWS alert the pilot?**
- a. When the a/c is below the bottom of the glide slope
 - b. When the a/c is more than 3' below the glide slope
 - c. When the ILS indicator shown more than half a full fly down indication
 - d. When the ILS indicator shown more than half a full fly-up indication
- 33. Which mode of the GPWS will warn the pilot when he/she descends below 500 ft with gear still up?**
- a. Mode 1
 - b. Mode 2b
 - c. Mode 3
 - d. Mode 4a
- 34. Below what height will GPWS mode 1 warn the pilot of a high rate of descent?**
- a. 2500 ft
 - b. 1800 ft
 - c. 790 ft
 - d. 500 ft
- 35. Regulations require a pilot to level his wings and initiate a maximum gradient climb to safe altitude under certain conditions. What are these conditions?**
- a. Any time he receives a GPWS warning
 - b. Any time he receives a GPWS alert
 - c. If he receives a GPWS alert, unless in VMC and in no danger
 - d. If he receives a GPWS warning unless in VMC and in no danger

- 36. If GPWS is found unserviceable, it must be repaired as soon as possible. If the FBO at destination has not got the necessary facilities, what must the pilot do?**
- He must divert to an airfield with repair facilities
 - He may land at destination, but must wait for the GPWS to be repaired
 - He may land at the destination, but must make the next landing at repair base
 - He may continue with his planned schedule until he lands at a repair base
- 37. How does a pilot know his GPWS is unserviceable?**
- ATC will tell him
 - A warning light will illuminate when he descends below 2500 ft
 - He will test it before flight
 - He has no means of knowing
- 38. Which of the following circumstances will GPWS not give warning about?**
- A high rate of descent close to the ground
 - Rising ground ahead
 - Approaching with the undercarriage not locked down
 - Approaching without flaps lowered
- 39. What is the principle of EGPWS?**
- It combines the radio altimeter with many sensor to descent rates at varying heights.
 - It sends radar signals ahead of the a/c to sense high round ahead
 - It gives an infra-red TV picture to the flight path through cloud
 - It used GPS and a map database to compute safe clearance from the ground ahead.

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13
B	C	A	B	B	A	B	D	B	C	D	A	B

14	15	16	17	18	19	20	21	22	23	24	25	26
A	D	A	D	D	C	B	B	A	D	A	A	D

27	28	29	30	31	32	33	34	35	36	37	38	39
B	A	B	B	A	D	D	A	D	D	C	B	D

SSR QUESTIONS

1. The special SSR codes are as follows: emergency _____, radio failure _____, entering airspace _____, unlawful interference with the conduct of the flight _____.
 - a. 7700; 7600; 2000; 7500.
 - b. 7700; 7600; 7500; 2000.
 - c. 7600; 7500; 2000; 7700.
 - d. 7500; 7600; 2000; 7700.
2. Secondary Surveillance Radar is a form of _____ radar with type emissions operating in the _____ band.
 - a. primary pulse SHF
 - b. primary pulse UHF
 - c. secondary FM SHF
 - d. secondary pulse UHF
3. If the SSR transponder IDENT button is pressed
 - a. it causes a momentary distinctive display to appear on the controller's screen.
 - b. an identification pulse is automatically and continuously transmitted for 2 seconds.
 - c. an identification pulse is automatically and continuously transmitted for 10 seconds.
 - d. an identification pulse is automatically and continuously transmitted for 20 seconds.
4. When using SSR the ground controller will ask the pilot to cancel mode C if there is a discrepancy of more than _____ feet between the altitude detected by the radar from the reply pulses and the altitude reported by the pilot read from an altitude with the subscale set to _____.
 - a. 100 ft Regional QNH
 - b. 300 ft 1013 HPa
 - c. 50 ft 1013 HPa
 - d. 300 ft Regional QNH
5. The SSR code which is appropriate for a FIR (inbound) crossing, where no other "squawk" has been allocated is:
 - a. 7500
 - b. 7600
 - c. 7700
 - d. 2000
6. Secondary radars require:
 - a. a target which will respond to the interrogation, and this target will always be an aircraft.
 - b. a target which will respond to the interrogation, and this target will always be ground based.

- c. a target which will respond to the interrogation, and this target may be either an aircraft or a ground based transponder.
- d. a quiescent target.

Answers

1	2	3	4	5	6
A	D	D	B	D	C

DME QUESTIONS

1. **Airborne DME equipment is able to discriminate between pulses intended for itself and pulses intended for other aircraft because:**
 - a. aircraft transmit and receive on different frequencies.
 - b. aircraft will only accept unique twin pulses.
 - c. aircraft reject pulses not synchronised with its own random pulse recurrence rate.
 - d. each aircraft has its own frequency allocation.
2. **A DME beacon having a transmit frequency of 962 MHz would have a receive frequency of:**
 - a. 1030 Mhz
 - b. 902 Mhz
 - c. 1025 Mhz
 - d. 962 Mhz
3. **A VOR/DME share the same first two letters of their respective identifiers; the last identifying letter of the DME is a Z. This means that:**
 - a. they are co-located.
 - b. they are more than 600m apart but serve the same location.
 - c. they are widely separated and do not serve the same location.
 - d. they are a maximum distance of 30m apart.
4. **Distance Measuring Equipment is an example of _____ radar operating on a frequency of _____ in the _____ band.**
 - a. primary 8800 MHz SHF
 - b. secondary 1030 MHz UHF
 - c. secondary 962 MHz UHF
 - d. primary 9375 MHz SHF
5. **A DME transponder does not respond to pulses received from radars other than DME because:**
 - a. each aircraft transmits pulses at a random rate.

- b. DME transmits and receives on different frequencies.
 - c. it will only accept the unique twin DME pulses.
 - d. DME only responds to the strongest 100 interrogators.
- 6. The range indicated by DME is considered to be accurate to within:**
- a. 3% of range
 - b. 1.25 % of range
 - c. 0.5 nm
 - d. +/-0.25 nm +/-1.25% of range.
- 7. A DME receiver is able to distinguish between replies to its own interrogations and replies to other aircraft because:**
- a. DME is secondary radar and each aircraft transmits and receives on a different frequency.
 - b. DME transponders reply to interrogations with twin pulses and the airborne equipment ejects all other pulses.
 - c. Each aircraft transmits pulses at a random rate and will only accept synchronized replies.
 - d. When DME is in the search mode it will only accept pulses giving the correct range.
- 8. When a DME transponder becomes saturated:**
- a. it reverts to standby.
 - b. it increases the number of pulse pairs to meet the demand.
 - c. it increases the receiver threshold to remove weaker signals.
 - d. it goes into a selective response mode of operation.
- 9. An aircraft flying at FL250 wishes to interrogate a DME beacon situated 400ft amsl. What is the maximum range likely to be achieved?**
- a. 210 nm
 - b. 198 nm
 - c. 175 nm
 - d. 222 nm
- 10. For a DME and a VOR to be said to be associated it is necessary for:**
- b. the DME to transmit on the same VHF frequency as the VOR.
 - b. the aerial separation not to exceed 100 feet in a TMA or 2000 feet outside a TMA.
 - c. the aerial separation not to exceed 100 metres in a TMA or 2000m outside a TMA.
 - d. both beacons to have the same first two letters for their ident' but the last letter of the DME to be a 'Z'.
- 11. The transmission frequency of a DME beacon is 63 MHz removed from the aircraft interrogator frequency to prevent:**
- a. interference from other radars.

- b. the airborne receiver locking on to primary returns from its own transmissions.
 - c. static interference.
 - d. receiver accepting replies intended for other interrogators.
- 12. For a VOR and a DME beacon to be said to be associated the aerial separation must not exceed _____ in a terminal area and _____ outside a terminal area.**
- a. 100 m 2000 m
 - b. 50 feet 200 feet
 - c. 30m 600m
 - d. 50 m 200 m
- 13. DME is a radar operating in the band and uses in order to obtain range information. The correct words to complete the above statement are:**
- a. primary SHF CW signals
 - b. secondary UHF twin pulses
 - c. secondary SHF "jittered pulses"
 - d. primary UHF pulse pairs
- 14. The receiver of airborne DME equipment is able to "lock on" to its own "reply pulses" because:**
- a. each aircraft has its own unique transmitter frequency and the receiver only accepts reply pulses having this frequency.
 - b. the reply pulses from the ground transmitter have the same frequency as the incoming interrogation pulses from the aircraft.
 - c. the aircraft receiver only accepts reply pulses which have the same time interval between successive pulses as the pulses being transmitted by its own transmitter.
 - d. the aircraft receiver only accepts reply pulses which arrive at a constant time interval.
- 15. DME operates in the frequency band, it transmits which give it the emission designator of.**
- a. SHF double size pulses PO1
 - b. UHF twin pulses PON
 - c. EHF twin pulses A9F
 - d. UHF double pulses J3E
- 16. Referring to DME during the initial stage of the "search" pattern before "lock-on":**
- a. the airborne receiver checks 150 pulses each second.
 - b. the airborne transmitter transmits 150 pulses each second.
 - c. the ground receiver maintains the ground transmitter pulse transmission rate at no more than 150 per second.

d. the aircraft transmits 24 pulses per second and the receiver checks a maximum of 150 pulses per second.

17. DME and VOR are “frequency paired” because:

- a. the same receiver can be used for both aids.
- b. the VOR transmitter is easily converted to the required DME frequency.
- c. cockpit workload is reduced.
- d. both ground transmitter aerials can be placed on the same site if required.

18. A DME receiver is able to distinguish between replies to its own interrogation pulses and those intended for other aircraft using the same transponder because:

- a. DME is a secondary radar and each aircraft transmits and receives on a different frequency.
- b. DME transponders reply to interrogations by means of twin pulses and the airborne equipment rejects all single pulses.
- c. each aircraft transmits pulses at a random rate(“jittering”) and will only accept replies that match this randomisation.
- d. when DME is in the range search mode it will accept only pulses separated by + or – 63 MHz from the interrogation frequency.

Answers

1	2	3	4	5	6	7	8	9
C	C	B	C	C	D	C	C	D

10	11	12	13	14	15	16	17	18
B	B	C	B	C	B	B	C	C

TCAS QUESTIONS

1. **On receipt of a TCAS RA your action is to:**
 - a. initiate the required manoeuvre immediately.
 - b. make a note of the details.
 - c. request a flight clearance deviation from ATC.
 - d. Do nothing until a TA is received.
2. **Which of the following statements concerning TCAS is correct:**
 - a. TCAS 2 provides avoidance instructions in the vertical and horizontal planes.
 - b. TCAS 2 cannot provide information on non-SSR equipped intruders.
 - c. TCAS 2 requires Mode S to be fitted to other aircraft.
 - d. TCAS 2 provides advice on which way to turn.
3. **With reference to Traffic Collision Avoidance Systems. The difference between TCAS I and II is that:**
 - a. TCAS II can provide Traffic Advisories and Resolution Advisories whilst TCAS I can only provide Traffic Advisories .
 - b. TCAS II can only be fitted to large aircraft which carry more than 30 passengers. Whilst TCAS I can be fitted to any aircraft.
 - c. TCAS I can be fitted to aircraft which carry transponders with Mode A only whilst TCAS II can only be fitted to aircraft whose transponders include either Mode C or Mode S.
 - d. TCAS II can only be fitted to aircraft which are equipped with EFIS.
4. **The aural messages provided by TCAS II are:**
 - a. Threat, Climb; Threat, Descend.
 - b. Climb left; Climb right; Descend left; Descend right.
 - c. Climb; Descend; Increase climb; Increase Descent.
 - d. Turn left, Turn Right, Increase Turn, Decrease Turn
5. **With reference to Traffic Collision Avoidance Systems:**
 - a. RAs may be disregarded only when the pilot visually identifies the potentially conflicting traffic and decides that no deviation is necessary and has the clearance confirmed by ATC.
 - b. RAs may be disregarded only when the pilot visually identifies the potentially conflicting traffic and decides that no deviation is necessary and has advised ATC of the other aircraft's proximity.
 - c. RAs must never be disregarded.
 - d. RAs may be disregarded only when

Answers

1	2	3	4	5
A	B	A	C	D

Questions

1. The accuracy required of a precision area navigation system is:
 - a. 0.25 nm
 - b. 2 nm
 - c. 1 nm
 - d. 0.5 nm

2. A basic 2D RNAV system will determine tracking information from:
 - a. twin DME
 - b. VOR/DME
 - c. Twin VOR
 - d. Any of the above

3. An aircraft using a basic 2D RNAV system is on a section between WP1 and WP2, a distance of 45 nm. The aircraft is 20nm from the phantom station, which is 270°/30 nm from the VOR/DME. The aircraft is 15nm from the VOR/DME. The range readout will show:
 - a. 15 nm
 - b. 20 nm
 - c. 25 nm
 - d. 30 nm

4. A 3D RNAV System indicates guidance capability in?
 - a. Horizontal plane only
 - b. Vertical plane only
 - c. Both Horizontal & Vertical Plane

Answers

1	2	3	4
C	B	B	C

EFIS Questions

1. **WXR display is controlled from:**
 - a. captains EHSI control only.
 - b. co-pilots EHSI control only.
 - c. a special control panel.
 - d. both captains and co-pilots EHSI control panels.
2. **Decision height is adjusted and set on the:**
 - a. flight management computer.
 - b. HSI section of the EFIS control panel.
 - c. ADI section of the EFIS control panel.
 - d. ADI or HSI
3. **WXR display is on:**
 - a. the captains CRT only.
 - b. the co-pilots CRT only.
 - c. a special screen.
 - d. on both the captains and co-pilots CRTs.
4. **Modes available for (EFIS) HSI on some units are:**
 - a. airspeed and Mach.
 - b. MAP and PLAN.
 - c. VOR, ILS, MAP and AUTO TRIM.
 - d. only from manometric sources.
5. **An EFIS as well as having a control panel, symbol generators and a remote light sensor also has:**
 - a. EADIS and EHSIs
 - b. EHSIs and altitude indicator
 - c. EADIs and EICAs
 - d. EADI and WXR display tubes

Answers

1	2	3	4	5
D	C	D	B	A

GPS QUESTIONS

1. **NAVSTAR/GPS operates in the band the receiver determines position by:**
 - a. UHF, range position lines
 - b. UHF, secondary radar principles
 - c. SHF, secondary radar principles
 - d. SHF, range position lines
2. **The orbital height and inclination of the NAVSTAR/GPS constellation are:**
 - a. 20180 km, 65°
 - b. 20180 km, 55°
 - c. 19099 km, 65°
 - d. 19099 km, 55°
3. **The model of the earth used for NAVSTAR/GPS is:**
 - a. WGS90
 - b. PZ90
 - c. WGS84
 - d. PZ84
4. **The minimum number of satellites required for a 3D fix is:**
 - a. 3
 - b. 4
 - c. 5
 - d. 6
5. **The NAVSTAR/GPS operational constellation comprises how many satellites**
 - a. 12
 - b. 21
 - c. 24
 - d. 30
6. **The most accurate fixing information will be obtained from:**
 - a. four satellites spaced 90° apart at 30° above the visual horizon
 - b. one satellite close to the horizon and 3 equally at 60° above the horizon
 - c. one satellite directly overhead and 3 equally spaced at 60° above the horizon
 - d. one satellite directly overhead and 3 spaced 120° apart close to the horizon
7. **The most significant error of GNSS is:**
 - a. GDOP
 - b. Receiver clock
 - c. Ionospheric propagation
 - d. Ephemeris
8. **The frequency available to non-authorized users of NAVSTAR/GPS is:**
 - a. 1227.6 MHz

- b. 1575.42 MHz
 - c. 1602 MHz
 - d. 1246 MHz
9. **The purpose of the pseudo-random noise codes in NAVSTAR/GPS is to:**
- a. identify the satellites
 - b. pass the almanac data
 - c. pass the navigation and system data
 - d. pass the ephemeris and time information
10. **The minimum number of satellites required for receiver autonomous integrity monitoring is:**
- a. 3
 - b. 4
 - c. 5
 - d. 6
11. **If a receiver has to download the almanac, the time to do this will be:**
- a. 2.5 minutes
 - b. 12.5 minutes
 - c. 25 minutes
 - d. 15 minutes
12. **The use of LAAS and WAAS remove the errors caused by:**
- a. propagation, selective availability, satellite ephemeris and clock
 - b. elective availability, satellite ephemeris and clock
 - c. GDOP, selective availability and propagation
 - d. receiver clock, GDOP, satellite ephemeris and clock
13. **The position derived from NAVSTAR/GPS satellites may be subject to the following errors:**
- a. selective availability, skywave interference, GDOP
 - b. propagation, selective availability, ephemeris
 - c. GDOP, static interference, instrument
 - d. ephemeris, GDOP, siting
14. **The PRN codes are used to:**
- a. determine the time interval between the satellite transmission and receipt of the signal at the receiver
 - b. pass ephemeris and clock data to the receivers
 - c. synchronise the receiver clocks with the satellites clocks
 - d. determine the range of the satellites from the receiver
15. **The availability of two frequencies in GNSS:**
- a. removes SV ephemeris and clock errors

- b. reduces propagation errors
- c. reduces errors caused by GDOP
- d. removes receiver clock errors

16. The NAVSTAR/GPS reference system is:

- a. A geo-centred 3D Cartesian co-ordinate system fixed with reference to the sun
- b. A geo-centred 3D Cartesian co-ordinate system fixed with reference to the prime meridian, equator and pole
- c. A geo-centred 3D Cartesian co-ordinate system fixed with reference to space
- d. A geo-centred 3D system based on latitude, longitude and altitude

17. The initial range calculation at the receiver is known as a pseudo-range, because it is not yet corrected for:

- a. receiver clock errors
- b. receiver and satellite clock errors
- c. receiver and satellite clock errors and propagation errors
- d. receiver and satellite clock errors and ephemeris errors

Answers

1	2	3	4	5	6	7	8	9
A	B	C	B	C	D	C	B	A

10	11	12	13	14	15	16	17
C	B	B	B	A	B	B	B

FMS Questions

1. **The sequence of displays accessed on initialisation is:**
 - a. POS INIT, IDENT, RTE
 - b. IDENT, RTE, POS INIT
 - c. IDENT, POS INIT, RTE
 - d. POS INIT, RTE, IDENT
2. **The FMC position will be at its most inaccurate:**
 - a. on take-off
 - b. at TOC
 - c. at TOD
 - d. on final approach
3. **The FMC navigational database can be accessed by the pilots:**
 - a. to update the database
 - b. to read information only
 - c. to change information between the 28 day updates
 - d. to change the information to meet the sector requirements
4. **Concerning FMC operation, which of the following is true:**
 - a. the FMC combines the long term accuracy of the IRS with the short term accuracy of the external reference
 - b. the FMC combines the long term accuracy of the IRS with the long term accuracy of the external reference
 - c. the FMC combines the short term accuracy of the IRS with the short term accuracy of the external reference
 - d. the FMC combines the short term accuracy of the IRS with the long term accuracy of the external reference
5. **The correct format for the input of position 50N 00527E to the CDU is:**
 - a. 5000.0N00527.0E
 - b. N50E00527
 - c. N5000.0E00527.0
 - d. N5000E00527
6. **The period of validity of the navigational database is:**
 - a. 28 days
 - b. 1 month
 - c. determined by the national authority and may be from 28 days to 91 days
 - d. 91 days

Answers

1	2	3	4	5	6
C	C	B	D	C	A